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Chapter 5
Synchronous Sequential Logic

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Chapter Objectives

- Know how to distinguish a sequential circuit from a combinational circuit.
- Understand the functionality of an *SR* latch, transparent latch (or *D* latch), *D* flip-flop, *JK* flip-flop, and *T* flip-flop.
- Know how to use the characteristic table and characteristic equation of a flip-flop.
- Know how to derive the state equation, state table, and state diagram of a clocked sequential circuit.
- Know the difference between Mealy and Moore finite state machines.
- Given the state diagram of a finite state machine, be able to write a HDL model of the machine.

Chapter Objectives

- Understand the HDL models of latches and flip-flops.
- Know how to write synthesizable HDL models of clocked sequential circuits.
- Know how to design a state machine using manual methods.
- Know how to eliminate equivalent states in a state table.
- Know how to define a one-hot state assignment code.
- Be able to design a sequential circuit with (a) *D* flip-flops, (b) *JK* flip-flops, and (c) *T* flip-flops.

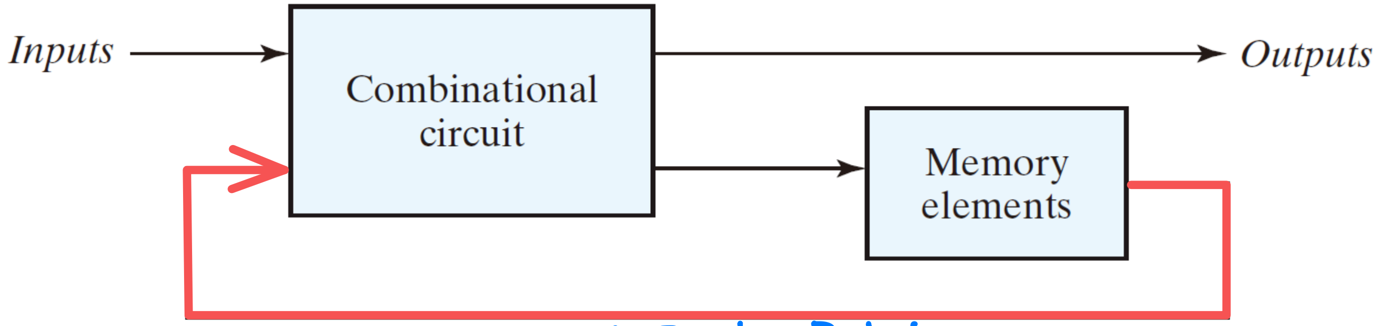
5.1 Introduction

- Combinational circuits
  - Contain no memory elements
  - The outputs depend only on the inputs

No memory Components Exisited.

5.2 Sequential Circuits (1/4)

- Sequential circuits



Feed Back Path!

FIGURE 5.1
Block diagram of sequential circuit

同步序項邏輯电路→時間到就處理
Synchronous Sequential Circuit

非同步序項邏輯电路→特定事件觸發就處理，不管時間！
Ansynchronous Sequential Circuit

  - A feedback path
  - The state of the sequential circuit
  - (inputs, current state) $\Rightarrow$ (outputs, next state)
  - Synchronous: the transition happens at discrete instants of time
  - Asynchronous: the transition happens at any instant of time

5.2 Sequential Circuits (2/4)

- Synchronous sequential circuits
  - Most commonly used
  - A master-clock generator to generate a periodic train of clock pulses 一个週期性的時脈
  - The clock pulses are distributed throughout the system
  - Clocked sequential circuits
  - No instability problems → 較穩定
  - The memory elements: flip-flops (正反器) → 一个正反器，就是一个Bit!
    - Binary cells capable of storing one bit of information
    - Two outputs: one for the normal value and one for the complement value → flip-flops 有兩个輸出，且这兩个輸出要互補。
    - Maintain a binary state indefinitely until directed by an input signal to switch states

5.2 Sequential Circuits (3/4)

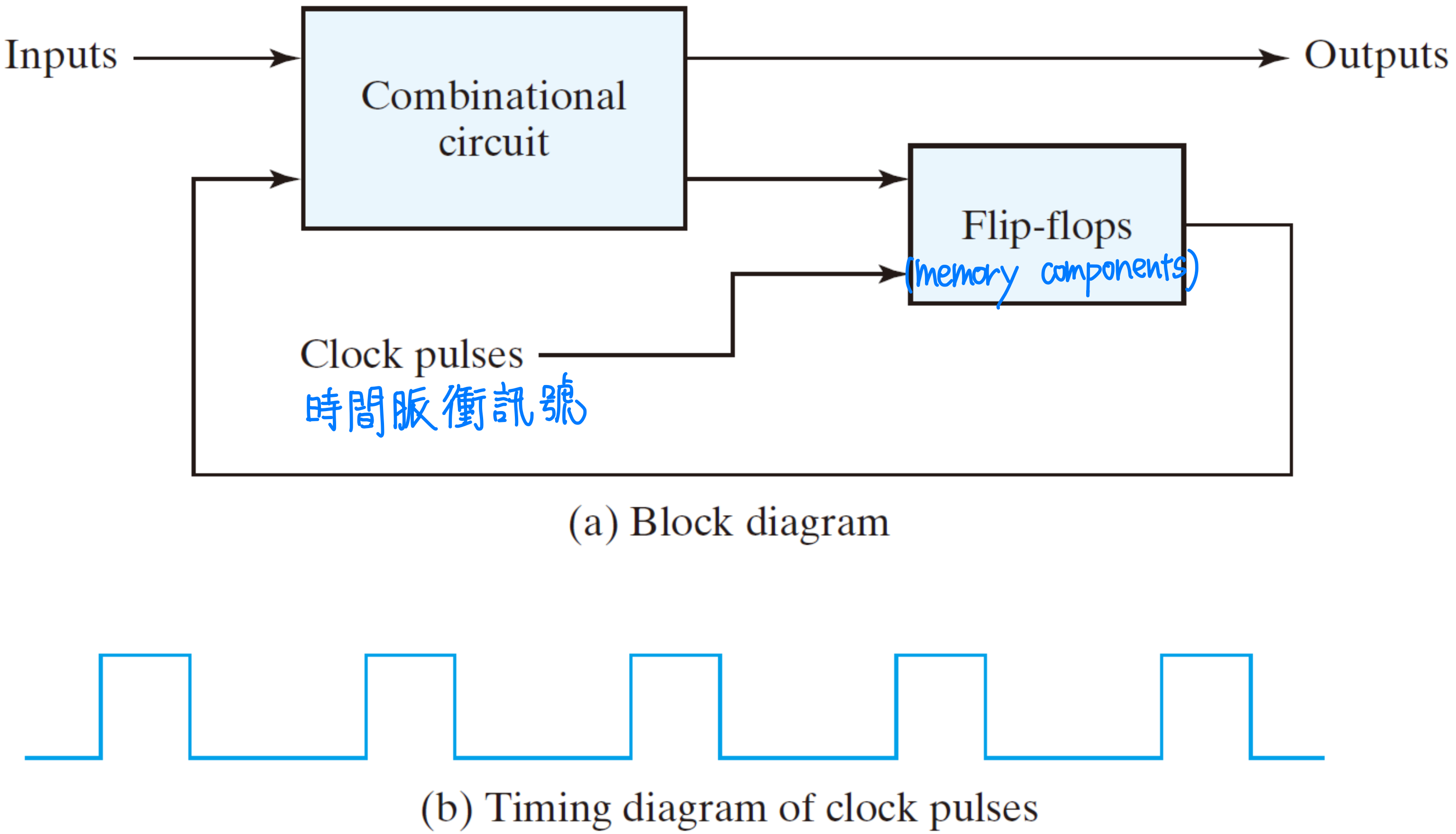


FIGURE 5.2
Synchronous clocked sequential circuit

5.2 Sequential Circuits (4/4)

- **Practice Exercise 5.1** Describe the fundamental difference between the output of a combinational circuit and the output of a sequential circuit.
  Answer:
  1. The output of a combinational circuit depends on only the inputs to the circuit.
  2. The output of a sequential circuit depends on the inputs to the circuit and the present state of the storage elements.

5.3 Latches (閂鎖器) (1/8)

ㄕㄨㄢ

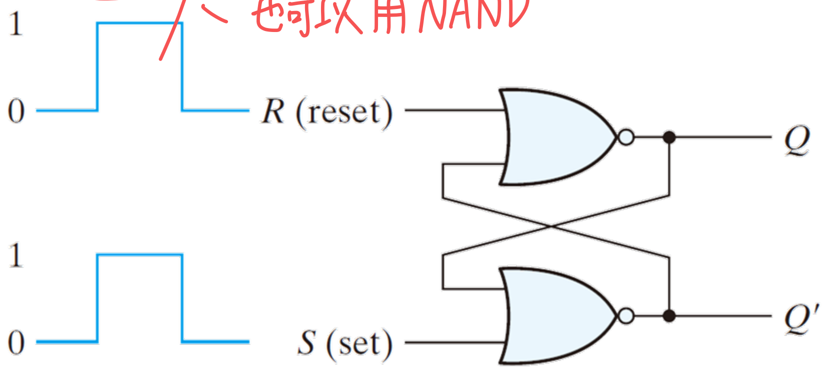
- Latch 閂鎖器
  - Storage element that operates with signal levels rather than signal transitions
  - Level-sensitive device
    (平坦区)
- Flip-flop 正反器
  - Constructed by the latches
  - Storage element that is controlled by a clock transition
  - Edge-sensitive device
    (上下衝的地方)

→ 都可以儲存訊號

→ flip-flop 根據 clock 來控制；Latch 根據準位來控制

5.3 Latches (2/8)
SR Latch

- Implementation approach #1: Two cross-coupled NOR gates
  交叉耦合
  、也可以用NAND
  - An asynchronous sequential circuit
  - Two inputs
    - *S* for set
    - *R* for reset
  - Two mutually complementary outputs Q and Q'
  - Functionalities
    - No operation
      - $(S, R) = (0, 0) \Rightarrow (Q(t+1), Q'(t+1)) = (Q(t), Q'(t))$
    - Reset
      - $(S, R) = (0, 1) \Rightarrow (Q(t+1), Q'(t+1)) = (0, 1)$
    - Set
      - $(S, R) = (1, 0) \Rightarrow (Q(t+1), Q'(t+1)) = (1, 0)$
    - Forbidden 輸出可預期，但不符合「輸出要互補」的規則！
      - $(S, R) = (1, 1) \Rightarrow (Q(t+1), Q'(t+1)) = (1, 0)$



(a) Logic diagram

NOR Truth Table

x	y	F
0	0	1
0	1	0
1	0	0
1	1	0

S、R為00時，輸出不變！
(與前一刻的輸出相同)

S	R	Q	Q'	
1	0	1	0	
√0	0	1	0	(after $S = 1, R = 0$)
0	1	0	1	
√0	0	0	1	(after $S = 0, R = 1$)
1	1	0	0	$\boxed{\text{(forbidden)}}$

輸出都是互補

操作被禁止！(沒有互補) →

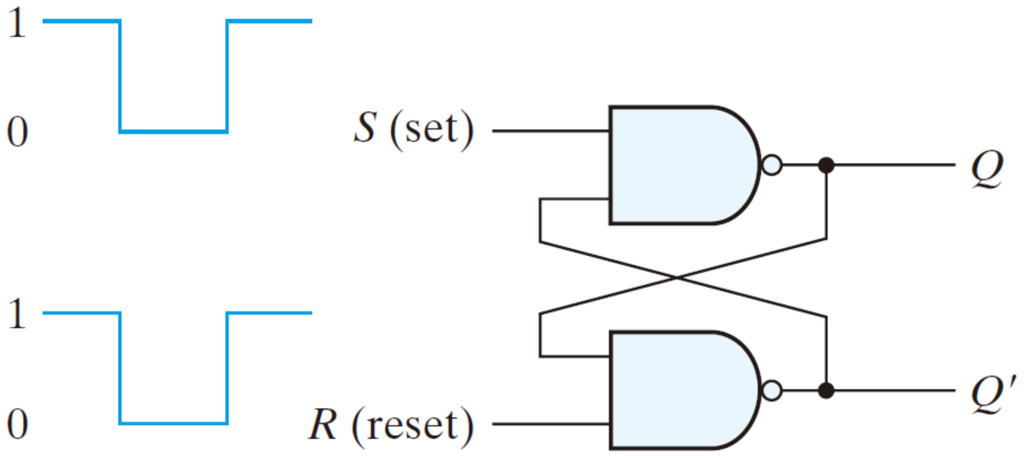
(b) Function table

FIGURE 5.3
***SR* latch with NOR gates**

S、R為1,0；0,1，則輸出為1,0；0,1。

5.3 Latches (3/8)
SR Latch

- Implementation approach #2: Two cross-coupled NAND gates



(a) Logic diagram

S	R	Q	Q'	
1	0	0	1	
1	1	0	1	(after $S = 1$, $R = 0$)
0	1	1	0	
1	1	1	0	(after $S = 0$, $R = 1$)
0	0	1	1	(forbidden)

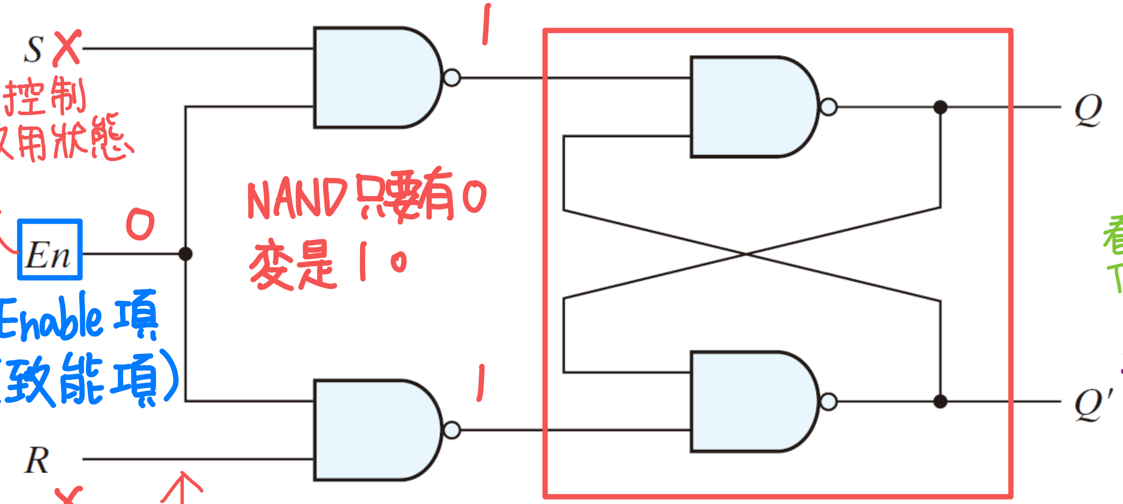
(b) Function table

FIGURE 5.4
***SR* latch with NAND gates**

5.3 Latches (4/8)
SR Latch

有 Control Input 的 Latch

- *SR* latch with control input
  - $En = 0$, no change
  - $En = 1$, NOR-gates-based *SR* latch



(a) Logic diagram

NAND Truth Tables

x	y	F
0	0	1
0	1	1
1	0	1
1	1	0

En	*S*	*R*	Next state of *Q*
0	X	X	No change → 輸出不变
1	0	0	No change
1	0	1	$Q = 0$; reset state
1	1	0	$Q = 1$; set state
1	1	1	$\boxed{\text{Indeterminate}}$ → 無法預期輸出

看NAND的 Truth Table →

看NOR的 Truth Table →

(b) Function table

FIGURE 5.5
***SR* latch with control input**

少了NOT的 D Latch

5.3 Latches (5/8)
SR Latch

- **Practice Exercise 5.2**

  (a) What input condition puts an *SR* NOR latch into an indeterminate state?

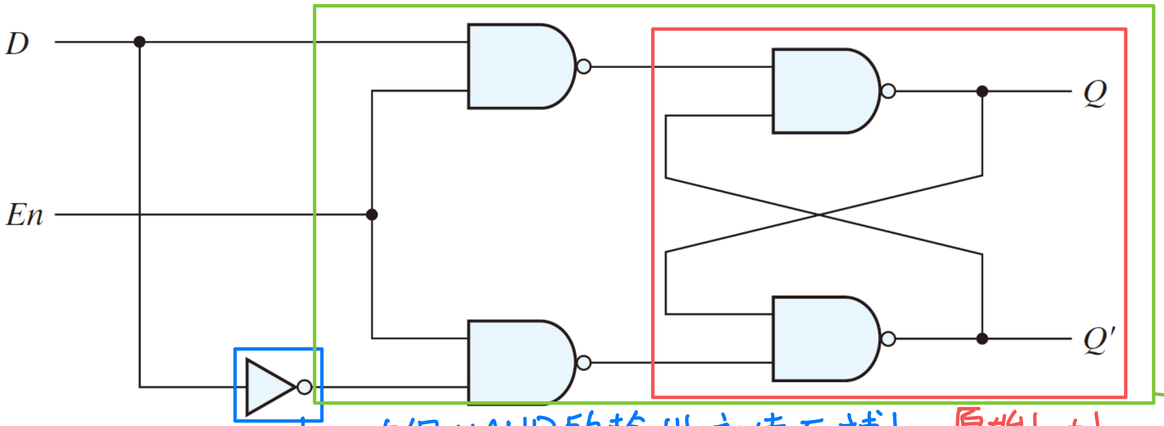
  Answer: Both inputs are 1.

  (b) What input condition puts an *SR* NAND latch into an indeterminate state?

  Answer: Both inputs are 0.

5.3 Latches (6/8)
D Latch

- *D* Latch (Transparent Latch)
  - Eliminate the undesirable conditions of the indeterminate state in the *SR* latch
  - *D* for data
  - Gated *D*-latch
  - $D \Rightarrow Q$ (or $Q = D$) when $En = 1$; no change (or $Q(t+1) = Q(t)$) when $En = 0$



→帶Enable項的Latch

→確保NAND的輸出永遠互補! 原始Latch

(a) Logic diagram

En	*D*	Next state of *Q*
0	X	No change
1	0	$Q = 0$; reset state
1	1	$Q = 1$; set state

輸入必等於輸出

(b) Function table

FIGURE 5.6
D latch

5.3 Latches (7/8)
D Latch

- Graphic Symbols

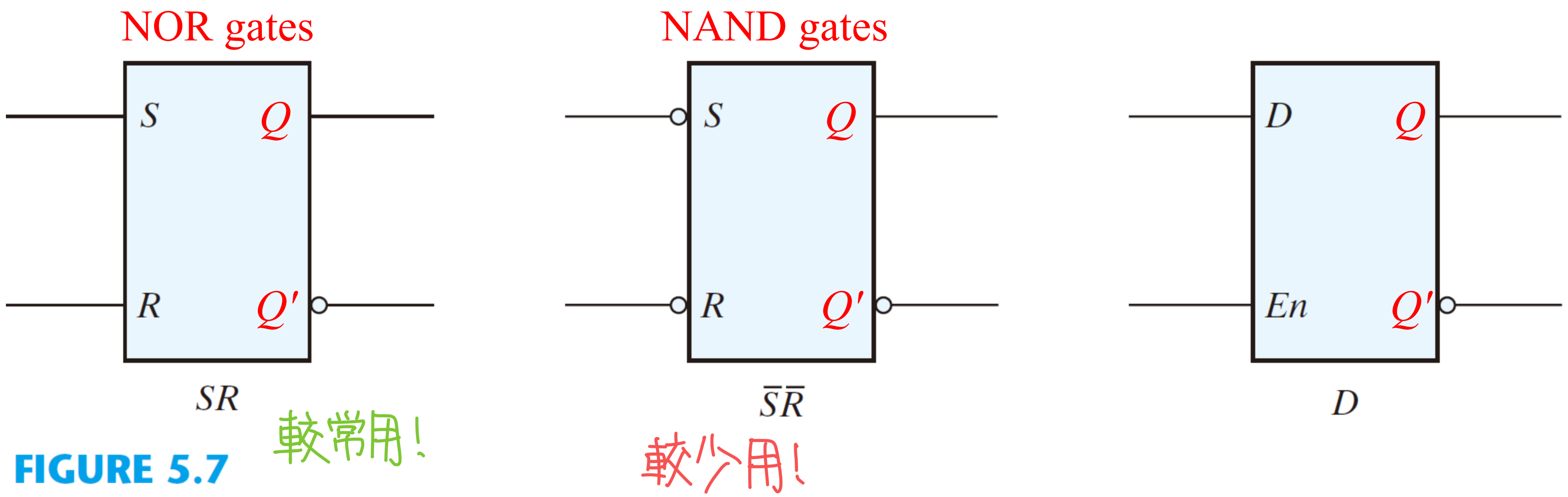


FIGURE 5.7
Graphic symbols for latches

5.3 Latches (8/8)
D Latch

- **Practice Exercise 5.3** Describe the functionality of a transparent latch.

  Answer:

  1. A transparent latch has a data input, an enable input, and two outputs.
  2. When the enable input is asserted, the output Q follows the input to the latch.
  3. When the enable input is de-asserted, the output Q is held at the value that was present at the moment the enable input was de-asserted.

5.4 Storage Element: Flip-Flops (1/16)

- A trigger (觸發) is a momentary change of the control input that causes a switch of the state of a latch or flip-flop.
  - Level-triggered – latches 閂鎖器
  - Edge-triggered – flip-flops 正反器

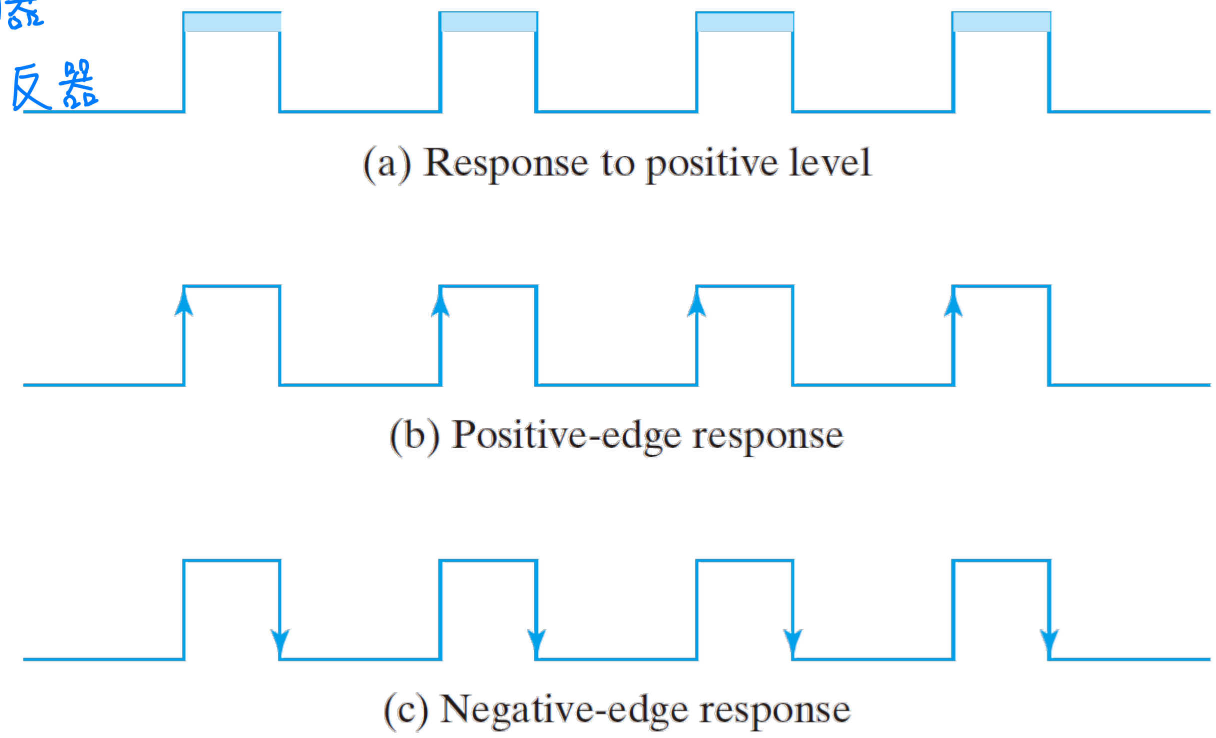


FIGURE 5.8
Clock response in latch and flip-flop

5.4 Storage Element: Flip-Flops (2/16)

準位觸發!

- If level-triggered flip-flops are used
  - the feedback path may cause instability problem

邊緣觸發

- Edge-triggered flip-flops
  - the state transition happens only at the edge
  - eliminate the multiple-transition problem

5.4 Storage Element: Flip-Flops (3/16)

Edge-Triggered *D* Flip-Flop 边緣触発型 D- flip flop

- Implementation approach #1: Leader-follower *D* flip-flop
  - Two separate *D* latches
  - A leader *D* latch (positive-level triggered)
  - A follower *D* latch (negative-level triggered)

两个 D Latch！

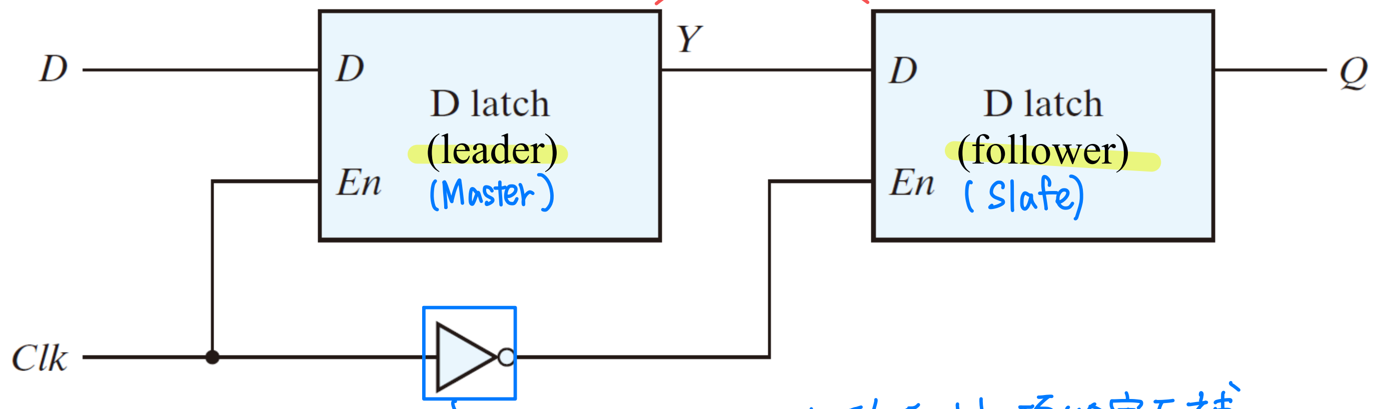
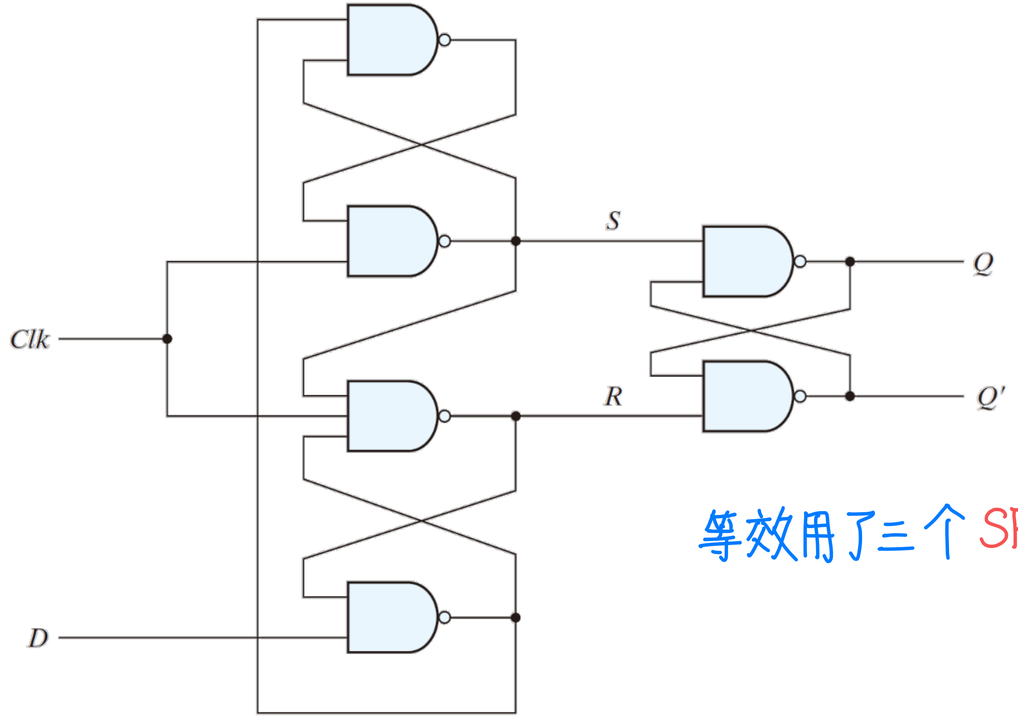


FIGURE 5.9
Leader-follower *D* flip-flop

导致这两个 Latch 的 Enable 项必定互補
→ 換言之，一次只会有一个 Latch 被触発！

5.4 Storage Element: Flip-Flops (4/16)
Edge-Triggered *D* Flip-Flop

- Implementation approach #2: Edge-triggered flip-flops
  - the state changes during a clock-pulse transition
- A *D*-type positive-edge-triggered flip-flop



等效用了三个 SR Latch

FIGURE 5.10
D-type positive-edge-triggered flip-flop

5.4 Storage Element: Flip-Flops (5/16)
Edge-Triggered *D* Flip-Flop

- Three basic *SR* latches
  - $(S, R) = (0, 1)$: $Q = 1$
  - $(S, R) = (1, 0)$: $Q = 0$
  - $(S, R) = (1, 1)$: no operation
  - $(S, R) = (0, 0)$: should be avoided

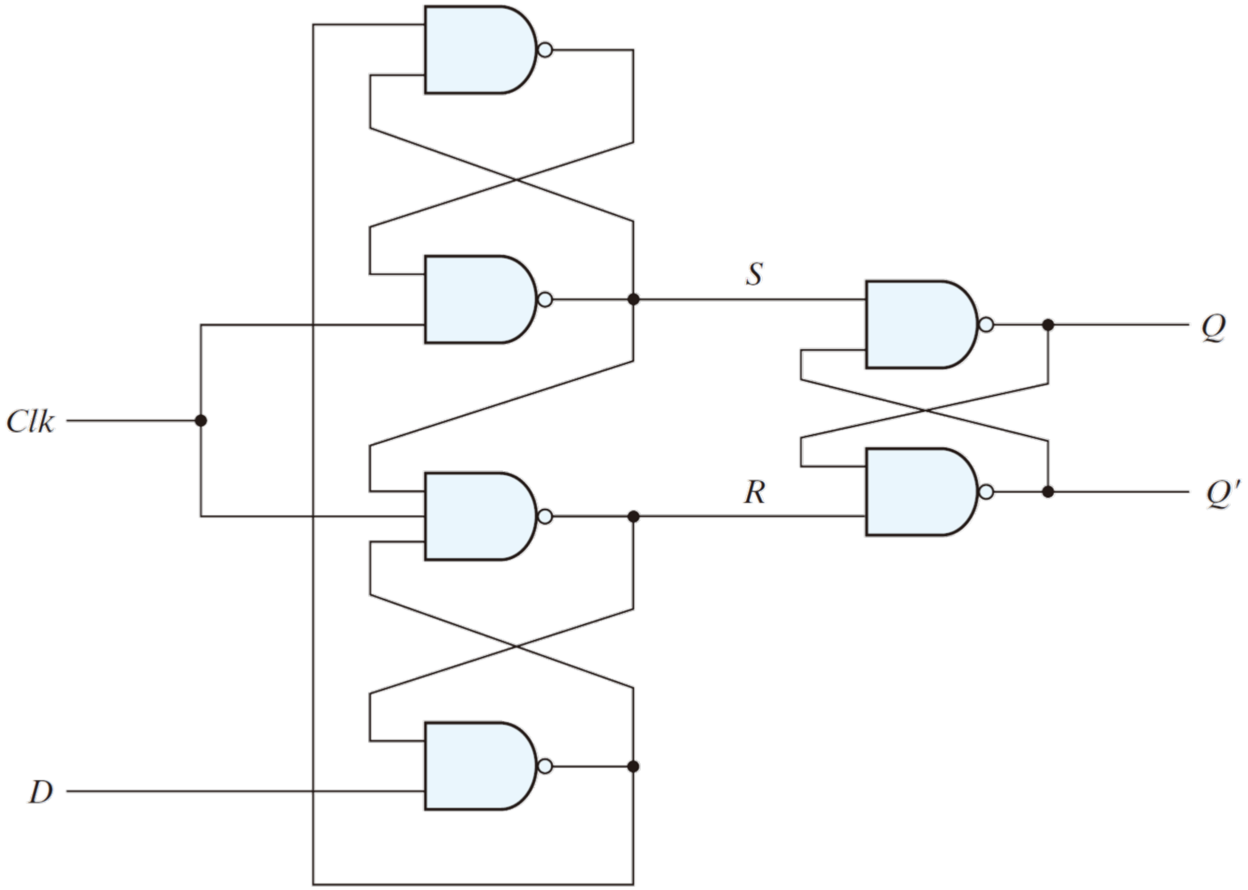


FIGURE 5.10
D-type positive-edge-triggered flip-flop

5.4 Storage Element: Flip-Flops (6/16)
Edge-Triggered *D* Flip-Flop

- The setup time
  - *D* input must be maintained at a constant value prior to the occurrence of the positive clock *Clk* pulse.
  - Data to the internal latches
- The hold time
  - *D* input must not change after the application of the positive *Clk* pulse
  - Clock to the internal latch
- The propagation delay time of the flip-flop is defined as the interval between the trigger edge and the stabilization of the output to a new state.

5.4 Storage Element: Flip-Flops (7/16)
Edge-Triggered *D* Flip-Flop

- Summary
  - $Clk = 0$: $(S, R) = (1, 1)$, no state change
  - $Clk = \uparrow$: state change once
  - $Clk = 1$: state holds
  - eliminate the feedback problems in sequential circuits
- All flip-flops must make their transition at the same time

5.4 Storage Element: Flip-Flops (8/16)
Edge-Triggered *D* Flip-Flop

- Graphic symbols
- The edge-triggered *D* flip-flops
  - The most economical and efficient
  - Positive-edge and negative-edge

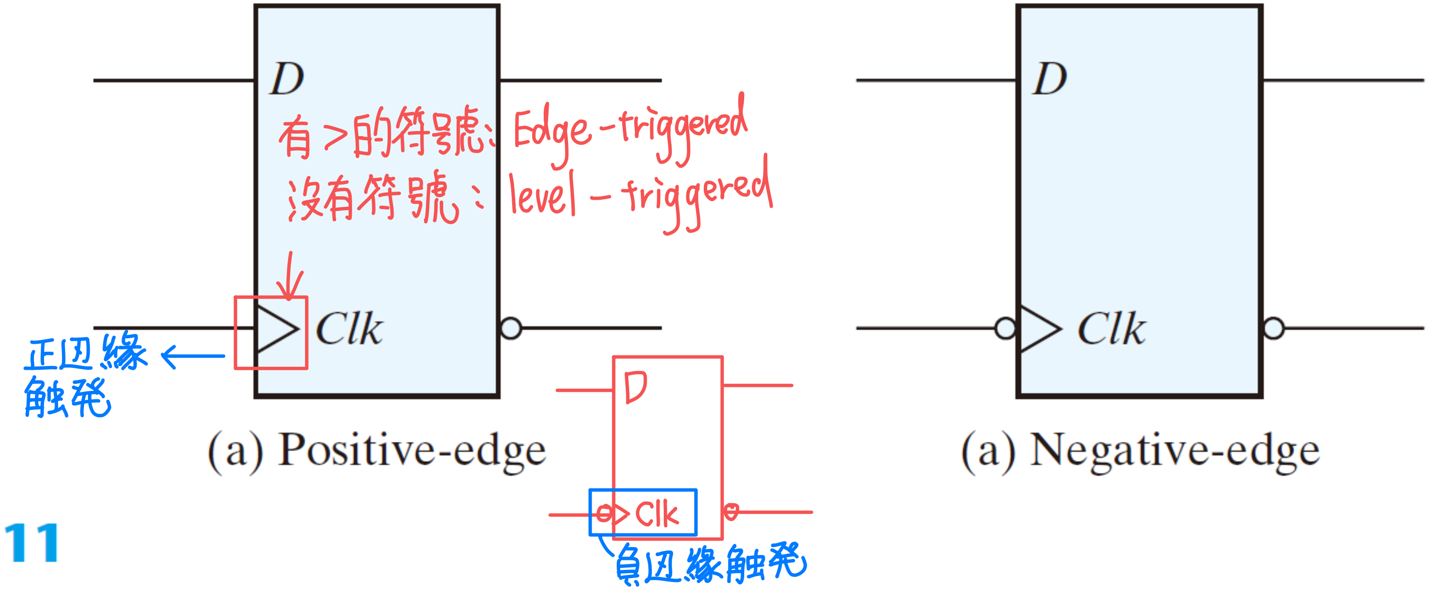


FIGURE 5.11
Graphic symbol for edge-triggered *D* flip-flop

5.4 Storage Element: Flip-Flops (9/16)
Edge-Triggered *D* Flip-Flop

- **Practice Exercise 5.4** What is meant by "a positive-edge flip-flop?"
  Answer: A positive-edge flip-flop is one that is activated by the rising (positive) edge of the clock (synchronizing signal).

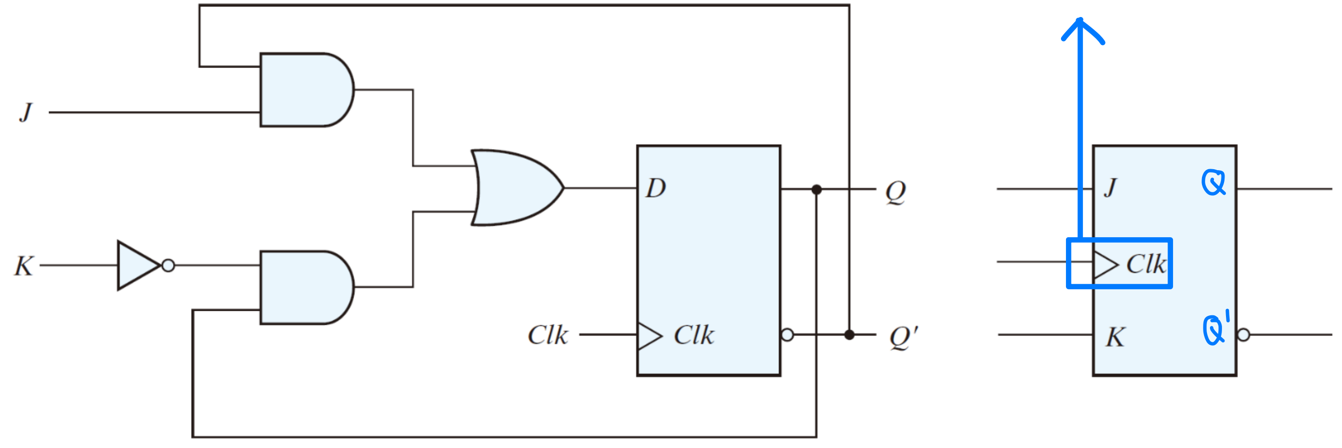
5.4 Storage Element: Flip-Flops (10/16)
Other Flip-Flops --- *JK* Flip-Flop JK正反器

(最泛用)

- $D = JQ'(t) + K'Q(t)$
  - $J = 0, K = 0: D = Q(t) \Rightarrow Q(t+1) = Q(t)$ (no change)
  - $J = 0, K = 1: D = 0 \Rightarrow Q(t+1) = 0$
  - $J = 1, K = 0: D = 1 \Rightarrow Q(t+1) = 1$
  - $J = 1, K = 1: D = Q'(t) \Rightarrow Q(t+1) = Q'(t)$

JK正反器的1,1 → 改变状態

正缘触発 JK正反器



(a) Circuit diagram (b) Graphic symbol

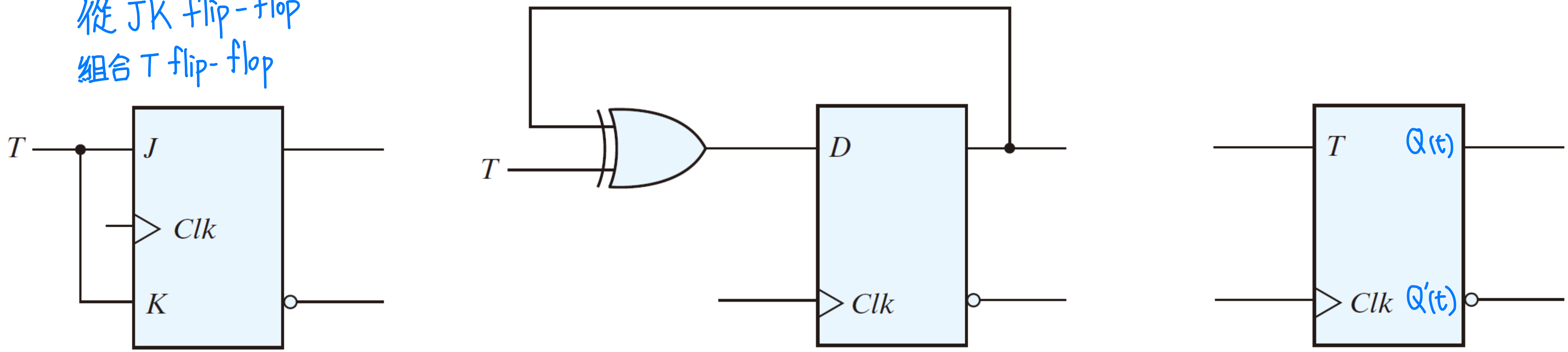
FIGURE 5.12
JK flip-flop

J	K	Q(t+1)
0	0	Q(t)
0	1	0
1	0	1
1	1	Q'(t)

5.4 Storage Element: Flip-Flops (11/16)
Other Flip-Flops ---*T* Flip-Flop T型正反器

- $D = T \oplus Q(t) = TQ'(t) + T'Q(t)$
  - $T = 0$: $D = Q(t) \Rightarrow Q(t+1) = Q(t)$ (no change)
  - $T = 1$: $D = Q'(t) \Rightarrow Q(t+1) = Q'(t)$

從 JK flip-flop
組合 T flip-flop



(a) From *JK* flip-flop

(b) From *D* flip-flop

利用 D-flipflop
組合 T-flipflop

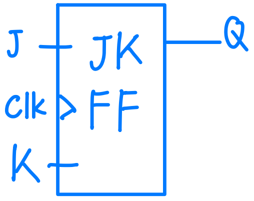
(c) Graphic symbol

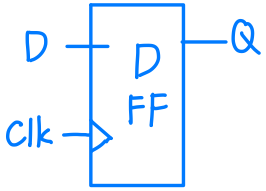
FIGURE 5.13
T flip-flop

5.4 Storage Element: Flip-Flops (12/16)
Other Flip-Flops

- Characteristic Tables (特徵表，特性表)

$D = JQ' + K'Q$





$D = TQ' + T'Q$
$= T \oplus Q$

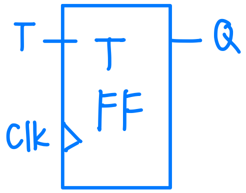


Table 5.1
Flip-Flop Characteristic Tables

***JK* Flip-Flop**

J	*K*	$Q(t+1)$	
0	0	$Q(t)$	No change
0	1	0	Reset
1	0	1	Set
1	1	$Q'(t)$	Complement

激勵表

$Q(t)$	$Q(t+1)$	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

***D* Flip-Flop**

D	$Q(t+1)$	
0	0	Reset
1	1	Set

***T* Flip-Flop**

T	$Q(t+1)$	
0	$Q(t)$	No change
1	$Q'(t)$	Complement

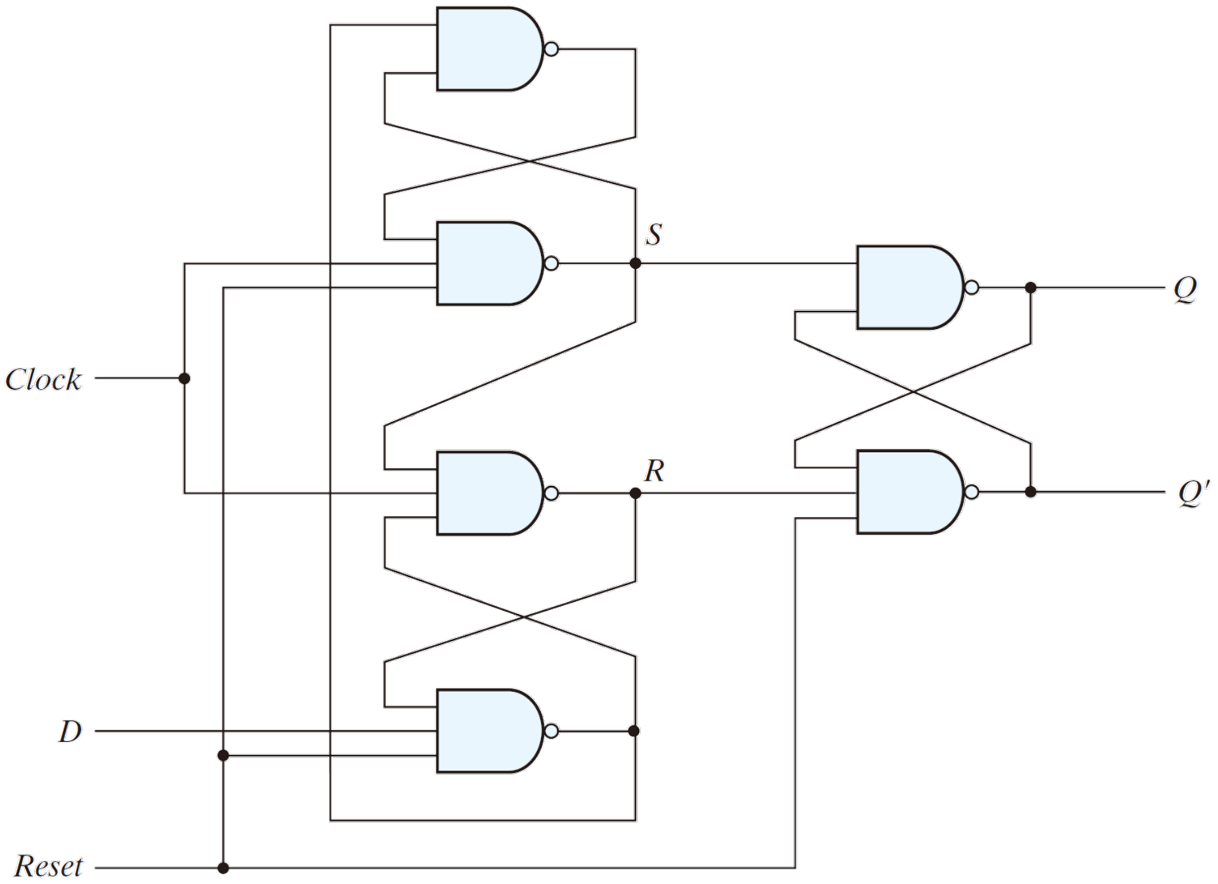
$Q(t)$	$Q(t+1)$	T
0	0	0
0	1	1
1	0	1
1	1	0

5.4 Storage Element: Flip-Flops (13/16)
Other Flip-Flops

- Characteristic Equations (特徵方程式、特性方程式)
  - *D* flip-flop
    - $Q(t+1) = D$
  - *JK* flip-flop
    - $Q(t+1) = JQ' + K'Q$
  - *T* flop-flop
    - $Q(t+1) = T \oplus Q$

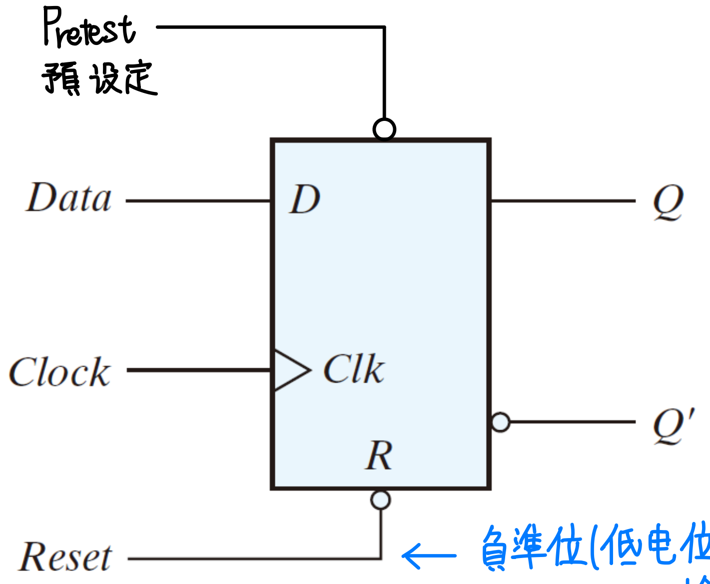
5.4 Storage Element: Flip-Flops (14/16) Direct Inputs

- Asynchronous *Set* and/or asynchronous *Reset*



(a) Circuit diagram

5.4 Storage Element: Flip-Flops (15/16)
Direct Inputs



(b) Graphic symbol

← 負準位(低电位)時，Reset 條件成立，輸出為0。

R	Clk	D	Q	Q'
0	X	X	0	1
1	↑	0	0	1
1	↑	1	1	0

(b) Function table

FIGURE 5.14
D flip-flop with asynchronous reset

Pretest、Rest 不與 Clk 同步！
在电路中伴演非同步设定的角色

5.4 Storage Element: Flip-Flops (16/16)
Direct Inputs

- **Practice Exercise 5.5** Describe the functionality of a *D*-type flip-flop.

  Answer:

  1. A *D*-type flip-flop has a *D* (data) input, a clock input, and possibly asynchronous or synchronous *reset* or *set* signal.
  2. If *set* or *reset* are not asserted, the clock signal synchronizes the transfer of *D* to the output *Q*.
  3. If *set* or *reset* are asynchronous, their action controls the flip-flop independently of the clock. *set* causes the output to be 1; *reset* causes the output to be 0.
  4. If *set* or *reset* are synchronous, their action has effect at the synchronizing edge of the clock.

5.5 Analysis of Clocked Sequential Circuits (1/33)
State Equations

- A sequential circuit
  - (inputs, current state) ⇒ (output, next state)
  - state transition table or **state table** 狀態表
  - state transition diagram or **state diagram** 狀態圖
- A **state equation** 狀態方程式 (also called a transition equation) specifies the next state of the sequential circuit as a function of the present state and input.
- An **output equation** specifies the output of the sequential circuit as a function of the present state and input.
- Design steps
  - Circuit diagram → Equations → State table → State diagram
  - State diagram → State table → Equations → Circuit diagram

狀態方程式：
描述记忆單元的狀態！
輸出方程式：
描述序項邏輯电路！
都跟「現態」、「輸入」有關！

5.5 Analysis of Clocked Sequential Circuits (2/33)
State Equations

Circuit → Equations → State table → State diagram

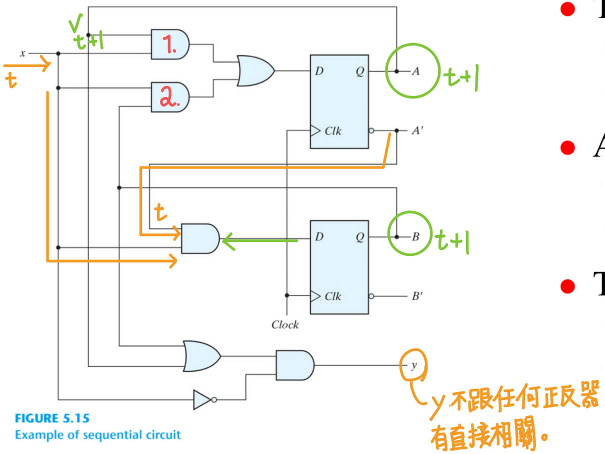


FIGURE 5.15
Example of sequential circuit

- The state equations 針對 flip-flop
  - $A(t+1) = \boxed{A(t)x(t)} + \boxed{B(t)x(t)}$ 1. 2.
  - $B(t+1) = A'(t)x(t)$
- A more compact form
  - $A(t+1) = Ax + Bx$
  - $B(t+1) = A'x$
- The output equation 針对 序項邏輯电路
  - $y(t) = [A(t) + B(t)]x'(t)$
  - $y = (A + B)x'$

State Table

Circuit → Equations → State table → State diagram

- The state equations
  - $A(t+1) = A(t)x(t) + B(t)x(t)$
  - $B(t+1) = A'(t)x(t)$
- A more compact form
  - $A(t+1) = Ax + Bx$
  - $B(t+1) = A'x$
- The output equation
  - $y(t) = [A(t) + B(t)]x'(t)$
  - $y = (A + B)x'$

$(Ax' + Bx')$

根據狀態方程式
決定 Next State

輸出方程式
決定 Output!

序項邏輯电路是可能沒有輸出.輸入的。

Table 5.2
State Table for the Circuit of Fig. 5.15

Present State		Input	Next State		Output
A	**B**	**x**	**A**	**B**	**y**
0	0	$\boxed{0}$	0	0	0
0	0	1	0	1	0
0	1	$\boxed{0}$	0	0	1
0	1	1	1	1	0
1	0	$\boxed{0}$	0	0	1
1	0	1	1	0	0
1	1	$\boxed{0}$	0	0	1
1	1	1	1	0	0

5.5 Analysis of Clocked Sequential Circuits (4/33)
State Diagram

Circuit → Equations → State table → State diagram

- A circle: a state ⟶描述「狀態」
- A directed line: the transition between the circles (or states).
  - Each directed line is labeled 'inputs/outputs'
  - Each line originates at a present state and terminates at a next state, depending on the input applied when the circuit is in the present state.

Table 5.3
Second Form of the State Table

輸入?

現在 →

Present State		Next State				Output	
		$x = 0$		$x = 1$		$x = 0$	$x = 1$
A	B	A	B	A	B	y	y
0	0	0	0	0	1	0	0
0	1	0	0	1	1	1	0
1	0	0	0	1	0	1	0
1	1	0	0	1	0	1	0

下一个狀態

2个記忆元件4圈圈
3个 …… 8圈圈

➔

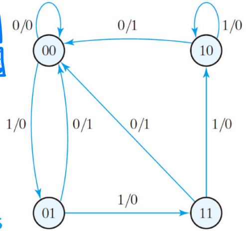


FIGURE 5.16
State diagram of the circuit of Fig. 5.15

5.5 Analysis of Clocked Sequential Circuits (5/33)
Flip-Flop Input Equations

- A sequential circuit can be fully described by the input (flip-flop-related) and output (logic gates-related) equations.
- Input equation: a set of Boolean functions to generate the inputs to the flip-flops.
  - Also called excitation functions (激勵方程式)
  - $D_A = Ax + Bx$
  - $D_B = A'x$
  - The symbols in input equations, D_A and D_B, imply the type of the flip-flop from the letter symbol
- Output equations: a set of Boolean functions to generate external outputs.
  - $y = (A + B)x'$

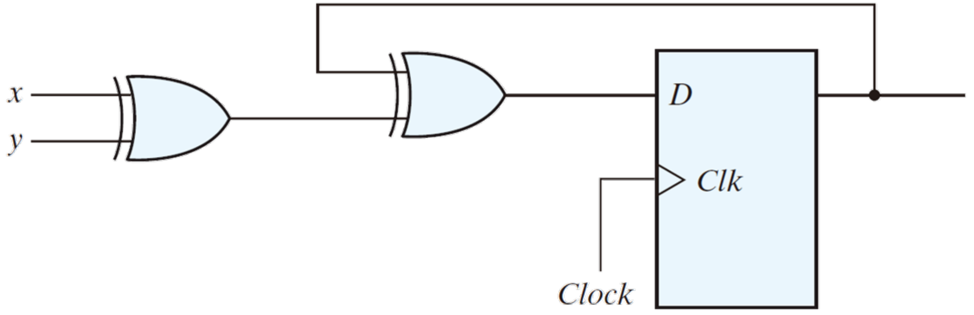
5.5 Analysis of Clocked Sequential Circuits (6/33)
Analysis with *D* Flip-Flops

<Review> D型正反器是「通透的正反器」!
輸入多少、輸出就是多少!

- The input equation
  - $D_A = A \oplus x \oplus y$

一樣!

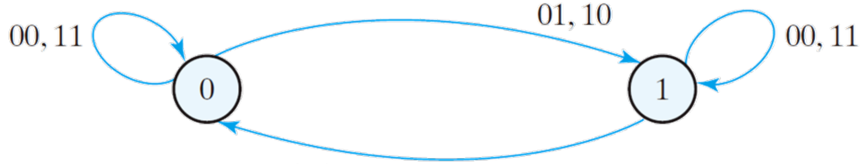
- The state equation
  - $A(t + 1) = A \oplus x \oplus y$
- Note that the input equation for *D* flip-flop is identical to the corresponding state equation.



(a) Circuit diagram

Present state	Inputs		Next state
A	x	y	A
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

(b) State table



(c) State diagram

FIGURE 5.17
Sequential circuit with *D* flip-flop

Analysis with *D* Flip-Flops

- **Practice Exercise 5.6** What determines the next state of a *D*-type flip-flop?

  Answer: The next state of a *D*-type flip-flop is the value of *D* at the synchronizing edge of the clock.

Analysis with *D* Flip-Flops: Example 1

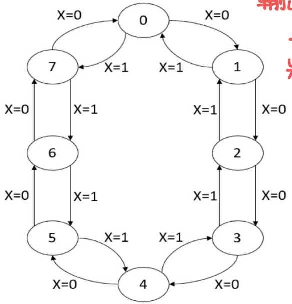
Problem → State diagram → State table → Equations → Circuit

- 設計具有3個D型正反器A, B, C與1個輸入X之3-bit counter。
  - $X = 0$時，電路會上數
  - $X = 1$時，電路會下數

3位元的上下數計數器！

3个记忆元件
$2^3=8$, 8圈圈

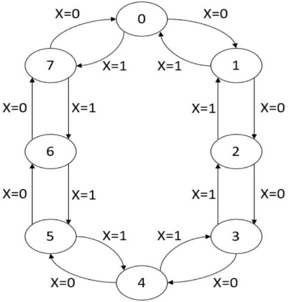
輸出？
三位元計數器的Bits狀態就是輸出！



State diagram

5.5 Analysis of Clocked Sequential Circuits (9/33)
Analysis with *D* Flip-Flops: Example 1

Problem → State diagram → State table → Equations → Circuit



State diagram

Present State			Input	Next State		
A	*B*	*C*	*X*	*A*	*B*	*C*
0	0	0	0	0	0	1
0	0	0	1	1	1	1
0	0	1	0	0	1	0
0	0	1	1	0	0	0
0	1	0	0	0	1	1
0	1	0	1	0	0	1
0	1	1	0	1	0	0
0	1	1	1	0	1	0
1	0	0	0	1	0	1
1	0	0	1	0	1	1
1	0	1	0	1	1	0
1	0	1	1	1	0	0
1	1	0	0	1	1	1
1	1	0	1	1	0	1
1	1	1	0	0	0	0
1	1	1	1	1	1	0

State table

5.5 Analysis of Clocked Sequential Circuits (10/33)
Analysis with *D* Flip-Flops: Example 1

Problem → State diagram → State table → Equations → Circuit

Present State			Input	Next State		
A	B	C	X	A	B	C
0	0	0	0	0	0	1
0	0	0	1	1	1	1
0	0	1	0	0	1	0
0	0	1	1	0	0	0
0	1	0	0	0	1	1
0	1	0	1	0	0	1
0	1	1	0	1	0	0
0	1	1	1	0	1	0
1	0	0	0	1	0	1
1	0	0	1	0	1	1
1	0	1	0	1	1	0
1	0	1	1	1	0	0
1	1	0	0	1	1	1
1	1	0	1	1	0	1
1	1	1	0	0	0	0
1	1	1	1	1	1	0

State table

AB \ CX	00	01	11	10
00		1		
01				1
11	1	1	1	
10	1		1	1

$$D_A = AC'X' + A'B'C'X + ABX + AB'C + A'BCX'$$

5.5 Analysis of Clocked Sequential Circuits (11/33)
Analysis with *D* Flip-Flops: Example 1

Problem → State diagram → State table → Equations → Circuit

Present State			Input	Next State		
A	B	C	X	A	B	C
0	0	0	0	0	0	1
0	0	0	1	1	1	1
0	0	1	0	0	1	0
0	0	1	1	0	0	0
0	1	0	0	0	1	1
0	1	0	1	0	0	1
0	1	1	0	1	0	0
0	1	1	1	0	1	0
1	0	0	0	1	0	1
1	0	0	1	0	1	1
1	0	1	0	1	1	0
1	0	1	1	1	0	0
1	1	0	0	1	1	1
1	1	0	1	1	0	1
1	1	1	0	0	0	0
1	1	1	1	1	1	0

State table

AB \ CX	00	01	11	10
00		1		1
01	1		1	
11	1		1	
10		1		1

$$D_B = BC'X' + B'C'X + BCX + B'CX'$$

5.5 Analysis of Clocked Sequential Circuits (12/33)
Analysis with *D* Flip-Flops: Example 1

Problem → State diagram → State table → Equations → Circuit

Present State			Input	Next State		
A	*B*	*C*	*X*	*A*	*B*	*C*
0	0	0	0	0	0	1
0	0	0	1	1	1	1
0	0	1	0	0	1	0
0	0	1	1	0	0	0
0	1	0	0	0	1	1
0	1	0	1	0	0	1
0	1	1	0	1	0	0
0	1	1	1	0	1	0
1	0	0	0	1	0	1
1	0	0	1	0	1	1
1	0	1	0	1	1	0
1	0	1	1	1	0	0
1	1	0	0	1	1	1
1	1	0	1	1	0	1
1	1	1	0	0	0	0
1	1	1	1	1	1	0

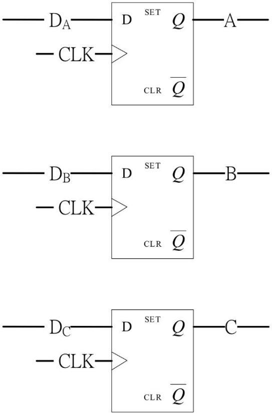
State table

AB \ *CX*	00	01	11	10
00	1	1		
01	1	1		
11	1	1		
10	1	1		

$D_C = C'$

5.5 Analysis of Clocked Sequential Circuits (13/33)
Analysis with *D* Flip-Flops: Example 1

Problem → State diagram → State table → Equations → Circuit



5.7 State Reduction and Assignment (1/9)
State Reduction

- Reductions on the number of flip-flops and the number of gates
- A reduction in the number of states may result in a reduction in the number of flip-flops
- An example state diagram

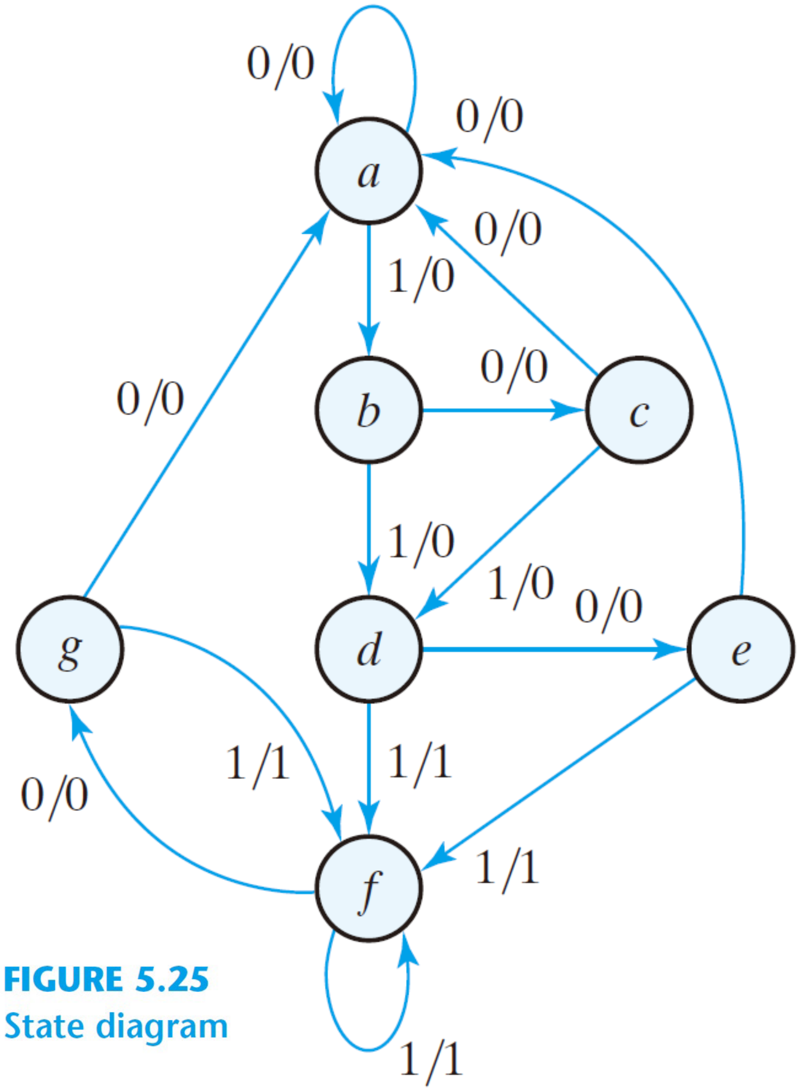


FIGURE 5.25
State diagram

5.7 State Reduction and Assignment (2/9)
State Reduction

state	***a***	***a***	***b***	***c***	***d***	***e***	***f***	***f***	***g***	***f***	***g***	***a***
input	0	1	0	1	0	1	1	0	1	0	0	
output	0	0	0	0	0	1	1	0	1	0	0	

- Only the input-output sequences are important
- Two circuits are equivalent
  - have identical outputs for all input sequences
  - the number of states is not important

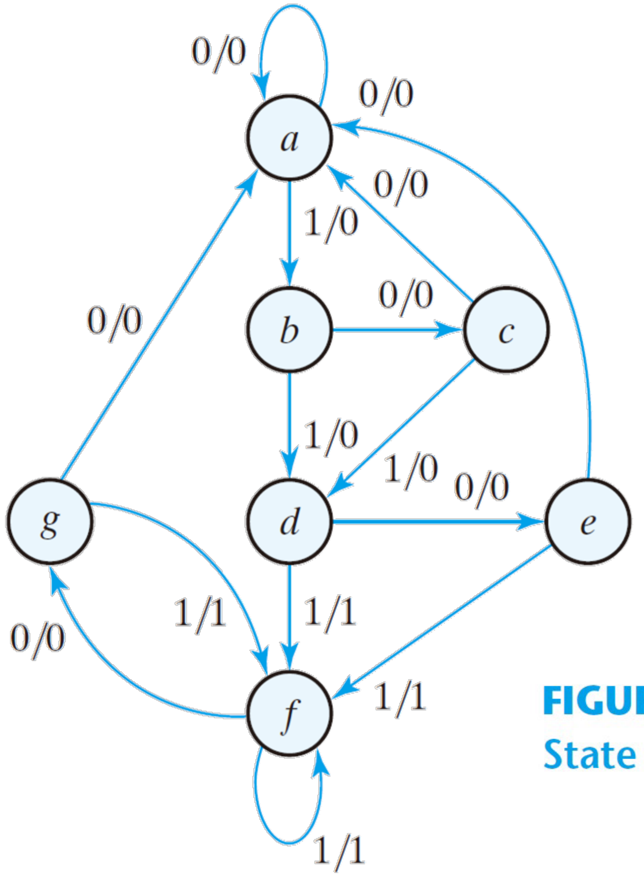


FIGURE 5.25
State diagram

5.7 State Reduction and Assignment (3/9)
State Reduction

- Equivalent states
  - two states are said to be equivalent
    - for each member of the set of inputs, they give exactly the same output and send the circuit to the same state or to an equivalent state
    - one of them can be removed

Table 5.6
State Table

Present State	Next State $x = 0$	Next State $x = 1$	Output $x = 0$	Output $x = 1$
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	f	0	1
e	a	f	0	1
f	g	f	0	1
g	a	f	0	1

5.7 State Reduction and Assignment (4/9)
State Reduction

Table 5.6
State Table

Present State	Next State $x = 0$	Next State $x = 1$	Output $x = 0$	Output $x = 1$
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	f	0	1
e	a	f	0	1
f	g	f	0	1
g	a	f	0	1

- Reducing the state table
  - $(a, b, c)(d, e, f, g)$ …. 根據輸出分成兩類

$\Rightarrow (a)(b, c)(d, e, f, g)$

$\Rightarrow (a)(b)(c)(d, e, f, g)$

$\Rightarrow (a)(b)(c)(d, f)(e, g)$

$\Rightarrow (a)(b)(c)(d\cancel{, f})(e\cancel{, g})$ …. (d, f)化簡成一個狀態, (e, g)化簡成一個狀態

$\Rightarrow (a)(b)(c)(d)(e)$

5.7 State Reduction and Assignment (5/9)
State Reduction

- Reducing the state table

(d, f)化簡成一個狀態 ，(e, g)化簡成一個狀態

Table 5.6
State Table

Present State	Next State $x = 0$	Next State $x = 1$	Output $x = 0$	Output $x = 1$
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	f	0	1
e	a	f	0	1
~~f~~	~~g~~	~~f~~	~~0~~	~~1~~
~~g~~	~~a~~	~~f~~	~~0~~	~~1~~

Table 5.7
Reducing the State Table

Present State	Next State $x = 0$	Next State $x = 1$	Output $x = 0$	Output $x = 1$
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	f	0	1
e	a	f	0	1
f	e	f	0	1

5.7 State Reduction and Assignment (6/9)
State Reduction

- The reduced finite state machine

Table 5.8
Reduced State Table

Present State	Next State $x=0$	Next State $x=1$	Output $x=0$	Output $x=1$
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	d	0	1
e	a	d	0	1

OLD

state	***a***	***a***	***b***	***c***	***d***	***e***	***f***	***f***	***g***	***f***	***g***	***a***
input	0	1	0	1	0	1	1	0	1	0	0	
output	0	0	0	0	0	1	1	0	1	0	0	

NEW

state	***a***	***a***	***b***	***c***	***d***	***e***	***d***	***d***	***e***	***d***	***e***	***a***
input	0	1	0	1	0	1	1	0	1	0	0	
output	0	0	0	0	0	1	1	0	1	0	0	

5.7 State Reduction and Assignment (7/9)
State Reduction

- The checking of each pair of states for possible equivalence can be done systematically
- The unused states are treated as don't-care condition ⇒ fewer combinational gates

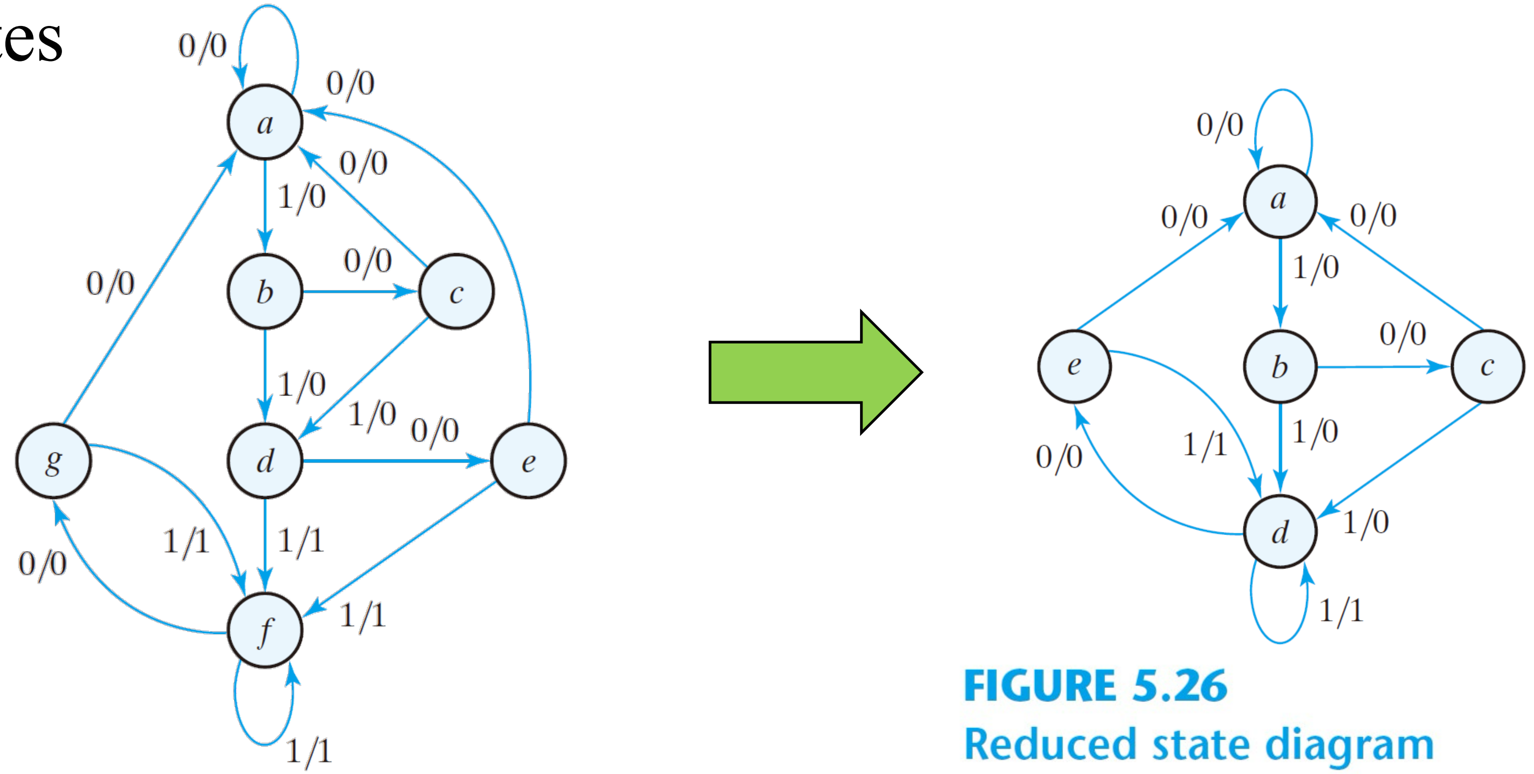


FIGURE 5.26
Reduced state diagram

5.7 State Reduction and Assignment (8/9)
State Assignment

- To minimize the cost of the combinational circuits
- Three possible binary state assignments

Table 5.9
Three Possible Binary State Assignments

State	Assignment 1, Binary	Assignment 2, Gray Code	Assignment 3, One-Hot
a	000	000	00001
b	001	001	00010
c	010	011	00100
d	011	010	01000
e	100	110	10000

5.7 State Reduction and Assignment (9/9)
State Assignment

- Any binary number assignment is satisfactory as long as each state is assigned a unique number
- Use binary assignment 1

Table 5.10
Reduced State Table with Binary Assignment 1

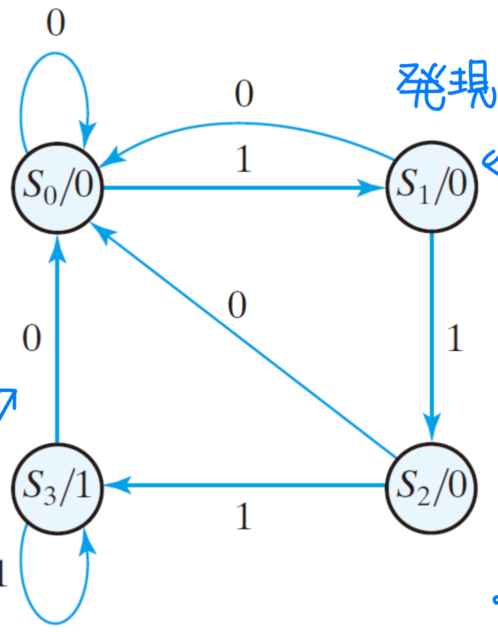
	Next State		Output	
Present State	$x = 0$	$x = 1$	$x = 0$	$x = 1$
000	000	001	0	0
001	010	011	0	0
010	000	011	0	0
011	100	011	0	1
100	000	011	0	1

5.8 Design Procedure (1/29)

- The word description of the circuit behavior (a state diagram)
- State reduction if necessary
- Assign binary values to the states
- Obtain the binary-coded state table
- Choose the type of flip-flops
- Derive the simplified flip-flop input equations and output equations
- Draw the logic diagram

5.8 Design Procedure (2/29)
Synthesis Using *D* Flip-Flops

- An example state diagram and state table, Moore's machine
  - if input sequence is "111", output y=1, otherwise, output y=0
  - 偵測序列"111"則輸出為1，否則輸出為0
  - In : 1111010…
  - Out: 0011000



發現一个1

第三个輸入是0
回到 S_0

第三个輸入是1
輸出1，回到 S_0

發現
連續2个1

FIGURE 5.27
State diagram for sequence detector

Table 5.11
State Table for Sequence Detector

Present State *A*	Present State *B*	Input *x*	Next State *A*	Next State *B*	Output *y*
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	0
0	1	1	1	0	0
1	0	0	0	0	0
1	0	1	1	1	0
1	1	0	0	0	1
1	1	1	1	1	1

5.8 Design Procedure (3/29)
Synthesis Using *D* Flip-Flops

Table 5.11
State Table for Sequence Detector

Present State		Input	Next State		Output
A	***B***	***x***	***A***	***B***	***y***
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	0
0	1	1	1	0	0
1	0	0	0	0	0
1	0	1	1	1	0
1	1	0	0	0	1
1	1	1	1	1	1

現在狀態 & 輸入　　下一狀態 & 輸出

- The flip-flop input equations
  - $A(t+1) = D_A(A, B, x) = \Sigma(3, 5, 7)$
  - $B(t+1) = D_B(A, B, x) = \Sigma(1, 5, 7)$
- The output equation
  - $y(A, B) = S(6, 7)$
- Logic minimization using the K map
  - $D_A = Ax + Bx$
  - $D_B = Ax + B'x$
  - $y = AB$

5.8 Design Procedure (4/28)
Sequence Detector --- Synthesis Using *D* Flip-Flops

優先選擇使用D型正反器！（最簡單 ^ᴗ^）

Problem → State diagram → State table → Equations → Circuit

Table 5.11
State Table for Sequence Detector

Present State		Input	Next State		Output
A	***B***	***x***	***A***	***B***	***y***
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	0
0	1	1	1	0	0
1	0	0	0	0	0
1	0	1	1	1	0
1	1	0	0	0	1
1	1	1	1	1	1

- The output equation
  - $y(A, B) = \Sigma(6, 7)$

D_A D_B D_C

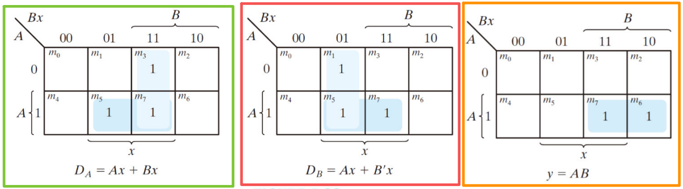


FIGURE 5.28
K-Maps for sequence detector

- The flip-flop input equations
  - $A(t + 1) = D_A(A, B, x) = \Sigma(3, 5, 7)$
  - $B(t + 1) = D_B(A, B, x) = \Sigma(1, 5, 7)$
- Logic minimization using the K-map
  - $D_A = Ax + Bx$
  - $D_B = Ax + B'x$
  - $y = AB$

5.8 Design Procedure (5/28)
Sequence Detector --- Synthesis Using *D* Flip-Flops

Problem → State diagram → State table → Equations → Circuit

- Logic minimization using the K-map
  - $D_A = Ax + Bx$
  - $D_B = Ax + B'x$
  - $y = AB$

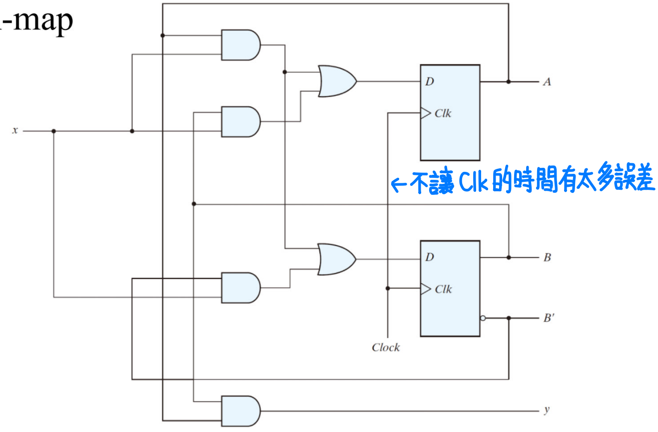
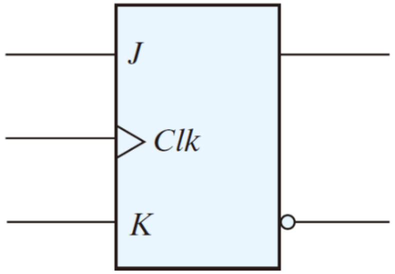


FIGURE 5.29
Logic diagram of a Moore-type sequence detector

5.8 Design Procedure (7/29)
Excitation Table of *JK* Flip-Flop

- A state diagram ⇒ flip-flop input functions
  - straightforward for *D* flip-flops



JK正反器
激勵表&
特徵表 要背!

Table 5.1
Flip-Flop Characteristic Tables

***JK* Flip-Flop**			
J	***K***	**Q(*t* + 1)**	
0	0	$Q(t)$	No change
0	1	0	Reset
1	0	1	Set
1	1	$Q'(t)$	Complement

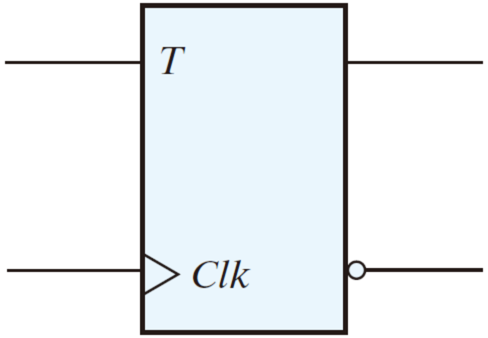
Table 5.12
Flip-Flop Excitation Tables

Q(*t*)	**Q(*t* = 1)**	***J***	***K***
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

(a) *JK* Flip-Flop

5.8 Design Procedure (8/29)
Excitation Table of *T* Flip-Flop

- A state diagram ⇒ flip-flop input functions



T型正反器
激勵表&
特徵表 要背!

Table 5.12
Flip-Flop Excitation Tables

$Q(t)$	$Q(t = 1)$	T
0	0	0
0	1	1
1	0	1
1	1	0

(b) *T* Flip-Flop

***T* Flip-Flop**

T	$Q(t+1)$	
0	$Q(t)$	No change
1	$Q'(t)$	Complement

5.8 Design Procedure (8/28)
Sequence Detector --- Synthesis Using JK Flip-Flops

Problem → State diagram → State table → Equations → Circuit

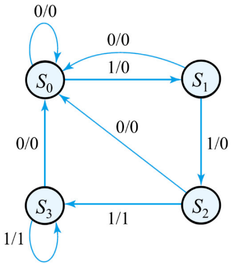


FIGURE 5.27
State diagram for sequence detector

Table 5.13
State Table and JK Flip-Flop Inputs

Present State		Input	Next State		Flip-Flop Inputs				Output
A	*B*	*x*	*A*	*B*	J_A	K_A	J_B	K_B	*y*
0	0	0	0	0	0	X	0	X	0
0	0	1	0	1	0	X	1	X	0
0	1	0	1	0	1	X	X	1	0
0	1	1	0	1	0	X	X	0	0
1	0	0	1	0	X	0	0	X	0
1	0	1	1	1	X	0	1	X	0
1	1	0	1	1	X	0	X	0	1
1	1	1	0	0	X	1	X	1	1

輸出只根暫存器狀態有關,和輸入無關!

JK正反器特徵表

Table 5.12
Flip-Flop Excitation Tables

$Q(t)$	$Q(t = 1)$	*J*	*K*
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

5.8 Design Procedure (9/28)
Sequence Detector --- Synthesis Using *JK* Flip-Flops

Problem → State diagram → State table → Equations → Circuit

Table 5.13
State Table and JK Flip-Flop Inputs

Present State		Input	Next State		Flip-Flop Inputs				Output
A	B	x	A	B	J_A	K_A	J_B	K_B	y
0	0	0	0	0	0	X	0	X	0
0	0	1	0	1	0	X	1	X	0
0	1	0	1	0	1	X	X	1	0
0	1	1	0	1	0	X	X	0	0
1	0	0	1	0	X	0	0	X	0
1	0	1	1	1	X	0	1	X	0
1	1	0	1	1	X	0	X	0	1
1	1	1	0	0	X	1	X	1	1

- The *JK* flip-flop inputs
  - $J_A(A, B, x) = \Sigma(2) + d(4, 5, 6, 7)$
  - $K_A(A, B, x) = \Sigma(7) + d(0, 1, 2, 3)$
  - $J_B(A, B, x) = \Sigma(1, 5) + d(2, 3, 6, 7)$
  - $K_B(A, B, x) = \Sigma(2, 7) + d(0, 1, 4, 5)$

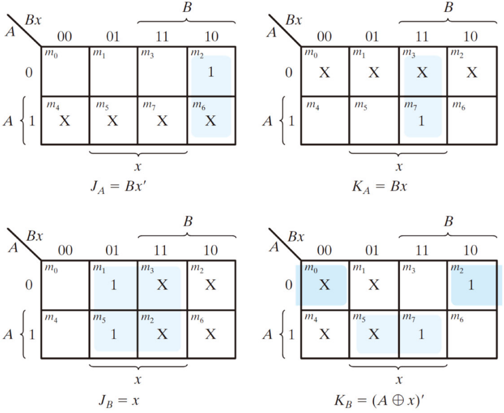


FIGURE 5.30
Maps for *J* and *K* input equations

5.8 Design Procedure (10/28)
Sequence Detector --- Synthesis Using *JK* Flip-Flops

Problem → State diagram → State table → Equations → Circuit

- $J_A = Bx'; K_A = Bx$
- $J_B = x; K_B = (A \oplus x)'$
- $y = AB$

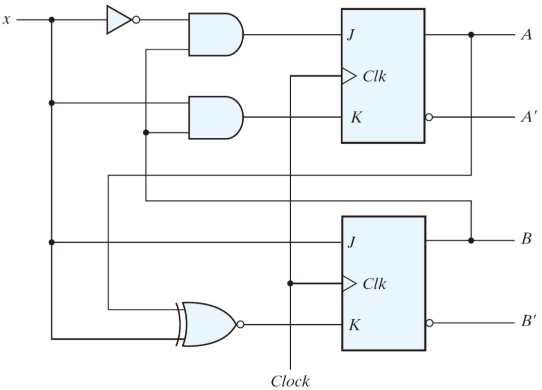


FIGURE 5.31
Logic diagram for sequential circuit with *JK* flip-flops

5.8 Design Procedure (11/29)
Synthesis Using *JK* Flip-Flops

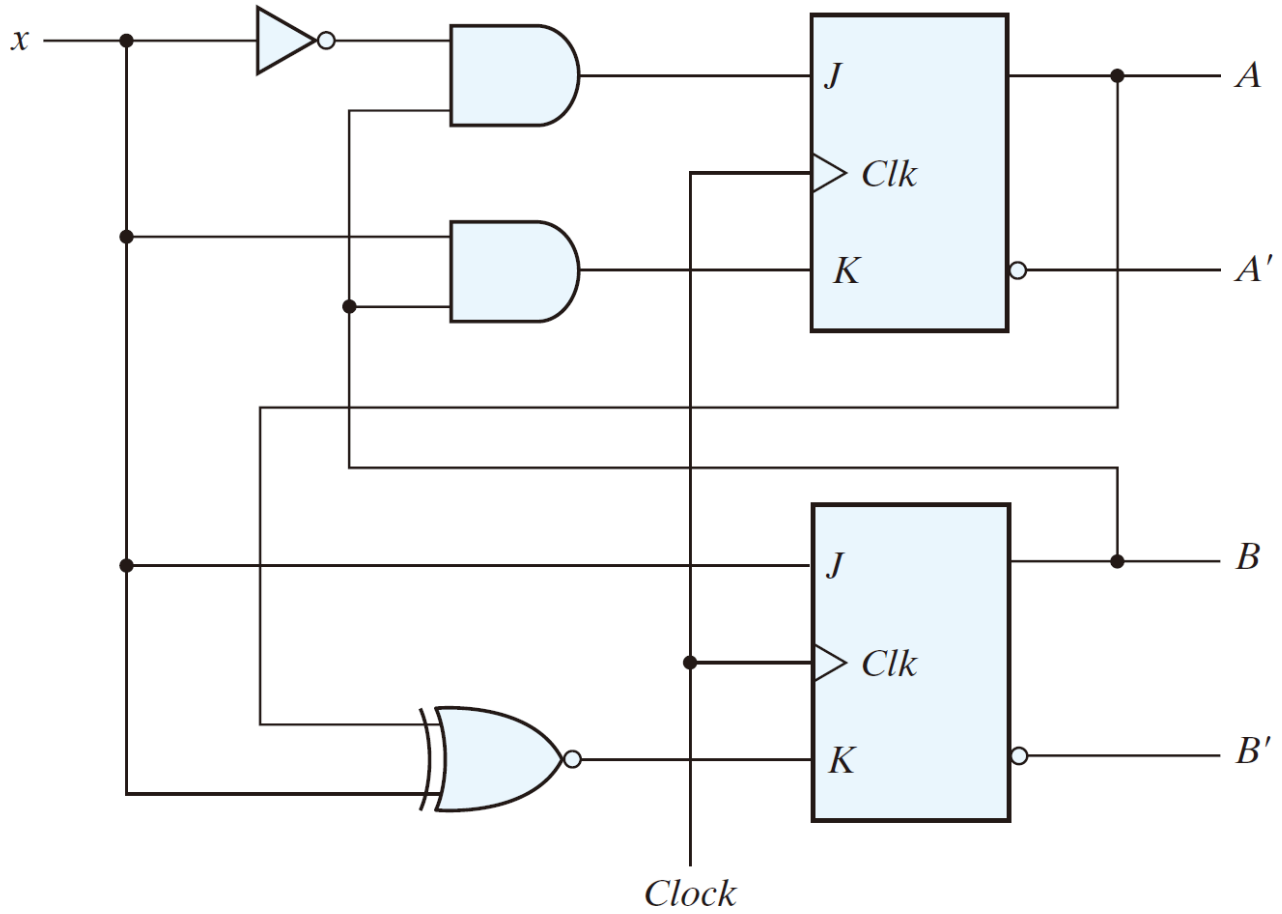


FIGURE 5.31
Logic diagram for sequential circuit with *JK* flip-flops

5.8 Design Procedure (12/29)
Synthesis Using *T* Flip-Flops

- A *n*-bit binary counter
  - the state diagram

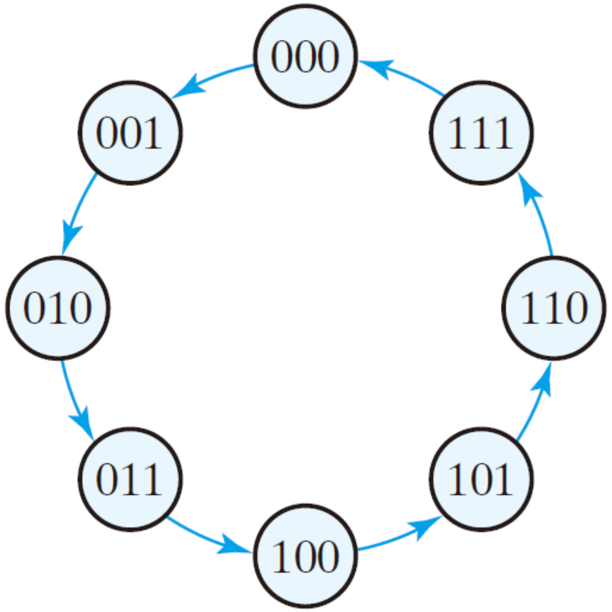
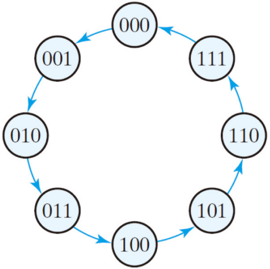


FIGURE 5.32
State diagram of three-bit binary counter

  - no inputs (except for the clock input)

5.8 Design Procedure (12/28)
Synthesis Using *T* Flip-Flops

Problem → State diagram → State table → Equations → Circuit



T型正反器特徵表

Table 5.12
Flip-Flop Excitation Tables

$Q(t)$	$Q(t=1)$	T
0	0	0
0	1	1
1	0	1
1	1	0

Table 5.14
State Table for Three-Bit Counter

Present State			Next State			Flip-Flop Inputs		
A_2	A_1	A_0	A_2	A_1	A_0	T_{A2}	T_{A1}	T_{A0}
0	0	0	0	0	1	0	0	1
0	0	1	0	1	0	0	1	1
0	1	0	0	1	1	0	0	1
0	1	1	1	0	0	1	1	1
1	0	0	1	0	1	0	0	1
1	0	1	1	1	0	0	1	1
1	1	0	1	1	1	0	1	1
1	1	1	0	0	0	1	1	1

Overflow!

→ 依特徵表寫狀態……

5.8 Design Procedure (14/29)
Synthesis Using *T* Flip-Flops

- The flip-flop inputs
  - $T_{A2}(A_2, A_1, A_0) = \Sigma(3, 7)$
  - $T_{A1}(A_2, A_1, A_0) = \Sigma(1, 3, 5, 7)$
  - $T_{A0}(A_2, A_1, A_0) = \Sigma(0, 1, 2, 3, 4, 5, 6, 7)$

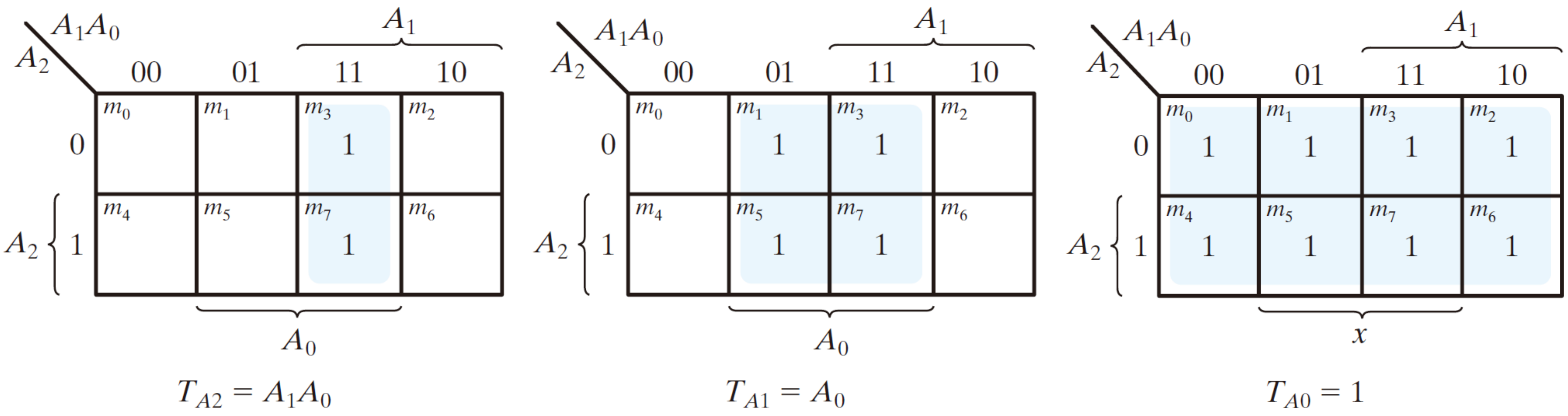


FIGURE 5.33
Maps for three-bit binary counter

5.8 Design Procedure (15/29)
Synthesis Using *T* Flip-Flops

- Logic simplification using the K map
  - $T_{A2} = A_1 A_2$
  - $T_{A1} = A_0$
  - $T_{A0} = 1$
- The logic diagram

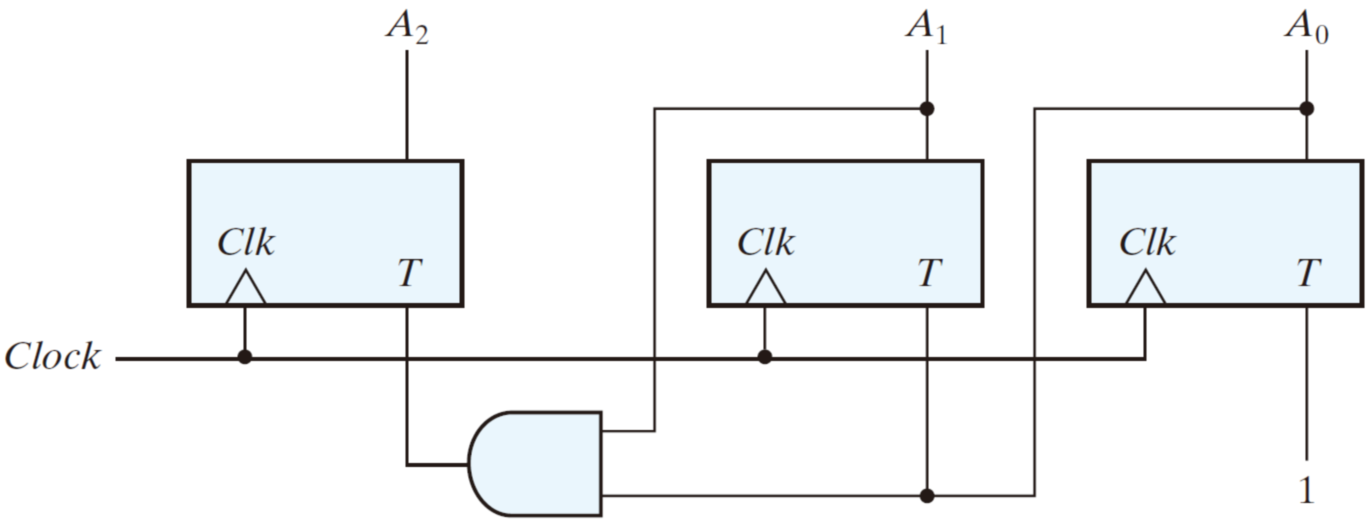


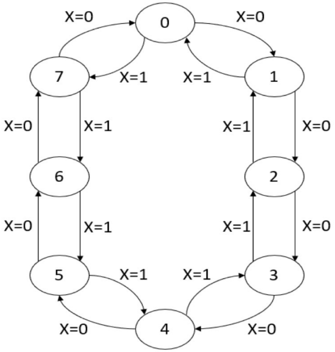
FIGURE 5.34
Logic diagram of three-bit binary counter

5.8 Design Procedure (15/28)
Analysis with *JK* Flip-Flops: Example 3

Problem → State diagram → State table → Equation → Circuit

- 設計具有3個*JK*型正反器*A*, *B*, *C*與1個輸入*X*之3-bit counter。
  - $X=0$時，電路會上數
  - $X=1$時，電路會下數

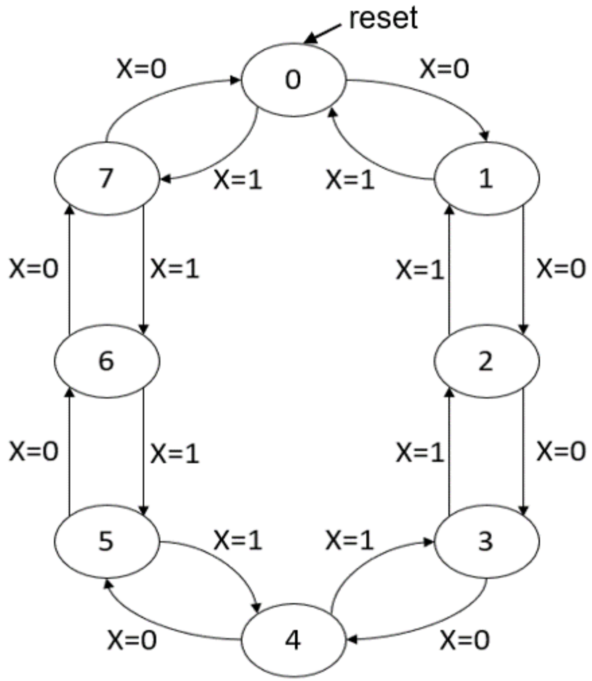
可以上下數!



State diagram

5.8 Design Procedure (17/29)
Analysis with *JK* Flip-Flops: Design an FSM Circuit (3)

Problem => State diagram => State table => Equation => Circuit



State diagram

Present State				Next State		
X	A	B	C	A	B	C
0	0	0	0	0	0	1
0	0	0	1	0	1	0
0	0	1	0	0	1	1
0	0	1	1	1	0	0
0	1	0	0	1	0	1
0	1	0	1	1	1	0
0	1	1	0	1	1	1
0	1	1	1	0	0	0
1	0	0	0	1	1	1
1	0	0	1	0	0	0
1	0	1	0	0	0	1
1	0	1	1	0	1	0
1	1	0	0	0	1	1
1	1	0	1	1	0	0
1	1	1	0	1	0	1
1	1	1	1	1	1	0

State table

5.8 Design Procedure (18/29)
Analysis with *JK* Flip-Flops: Design an FSM Circuit (3)

Problem => State diagram => State table => Equation => Circuit

Present State				Next State			Next State JK					
X	A	B	C	A	B	C	JA	KA	JB	KB	JC	KC
0	0	0	0	0	0	1	0	X	0	X	1	X
0	0	0	1	0	1	0	0	X	1	X	X	1
0	0	1	0	0	1	1	0	X	X	0	1	X
0	0	1	1	1	0	0	1	X	X	1	X	1
0	1	0	0	1	0	1	X	0	0	X	1	X
0	1	0	1	1	1	0	X	0	1	X	X	1
0	1	1	0	1	1	1	X	0	X	0	1	X
0	1	1	1	0	0	0	X	1	X	1	X	1
1	0	0	0	1	1	1	1	X	1	X	1	X
1	0	0	1	0	0	0	0	X	0	X	X	1
1	0	1	0	0	0	1	0	X	X	1	1	X
1	0	1	1	0	1	0	0	X	X	0	X	1
1	1	0	0	0	1	1	X	1	1	X	1	X
1	1	0	1	1	0	0	X	0	0	X	X	1
1	1	1	0	1	0	1	X	0	X	1	1	X
1	1	1	1	1	1	0	X	0	X	0	X	1

Table 5.12
Flip-Flop Excitation Tables

$Q(t)$	$Q(t = 1)$	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

(a) *JK* Flip-Flop

5.8 Design Procedure (18/28)
Analysis with *JK* Flip-Flops: Example 3

Problem → State diagram → State table → Equation → Circuit

Present State				Flip-Flop Inputs					
X	A	B	C	JA	KA	JB	KB	JC	KC
0	0	0	0	0	X	0	X	1	X
0	0	0	1	0	X	1	X	X	1
0	0	1	0	0	X	X	0	1	X
0	0	1	1	1	X	X	1	X	1
0	1	0	0	X	0	0	X	1	X
0	1	0	1	X	0	1	X	X	1
0	1	1	0	X	0	X	0	1	X
0	1	1	1	X	1	X	1	X	1
1	0	0	0	1	X	1	X	1	X
1	0	0	1	0	X	0	X	X	1
1	0	1	0	0	X	X	1	1	X
1	0	1	1	0	X	X	0	X	1
1	1	0	0	X	1	1	X	1	X
1	1	0	1	X	0	0	X	X	1
1	1	1	0	X	0	X	1	1	X
1	1	1	1	X	0	X	0	X	1

State table

XA \ BC	00	01	11	10
00			1	
01	X	X	X	X
11	X	X	X	X
10	1			

$J_A = XB'C' + X'BC$

JA = KA

XA \ BC	00	01	11	10
00	X	X	X	X
01			1	
11	1			
10	X	X	X	X

$K_A = XB'C' + X'BC$

5.8 Design Procedure (19/28)
Analysis with *JK* Flip-Flops: Example 3

Problem → State diagram → State table → Equation → Circuit

Present State				Flip-Flop Inputs					
X	A	B	C	JA	KA	JB	KB	JC	KC
0	0	0	0	0	X	0	X	1	X
0	0	0	1	0	X	1	X	X	1
0	0	1	0	0	X	X	0	1	X
0	0	1	1	1	X	X	1	X	1
0	1	0	0	X	0	0	X	1	X
0	1	0	1	X	0	1	X	X	1
0	1	1	0	X	0	X	0	1	X
0	1	1	1	X	1	X	1	X	1
1	0	0	0	1	X	1	X	1	X
1	0	0	1	0	X	0	X	X	1
1	0	1	0	0	X	X	1	1	X
1	0	1	1	0	X	X	0	X	1
1	1	0	0	X	1	1	X	1	X
1	1	0	1	X	0	0	X	X	1
1	1	1	0	X	0	X	1	1	X
1	1	1	1	X	0	X	0	X	1

State table

XA \ BC	00	01	11	10
00		1	X	X
01		1	X	X
11	1		X	X
10	1		X	X

$J_B = XC' + X'C$

$J_B = K_B$

XA \ BC	00	01	11	10
00	X	X	1	
01	X	X	1	
11	X	X		1
10	X	X		1

$K_B = XC' + X'C$

Analysis with *JK* Flip-Flops: Example 3

Problem → State diagram → State table → Equation → Circuit

Present State				Flip-Flop Inputs					
X	A	B	C	JA	KA	JB	KB	JC	KC
0	0	0	0	0	X	0	X	1	X
0	0	0	1	0	X	1	X	X	1
0	0	1	0	0	X	X	0	1	X
0	0	1	1	1	X	X	1	X	1
0	1	0	0	X	0	0	X	1	X
0	1	0	1	X	0	1	X	X	1
0	1	1	0	X	0	X	0	1	X
0	1	1	1	X	1	X	1	X	1
1	0	0	0	1	X	1	X	1	X
1	0	0	1	0	X	0	X	X	1
1	0	1	0	0	X	X	1	1	X
1	0	1	1	0	X	X	0	X	1
1	1	0	0	X	1	1	X	1	X
1	1	0	1	X	0	0	X	X	1
1	1	1	0	X	0	X	1	1	X
1	1	1	1	X	0	X	0	X	1

State table

XA \ BC	00	01	11	10
00	1	X	X	1
01	1	X	X	1
11	1	X	X	1
10	1	X	X	1

$J_C = 1$

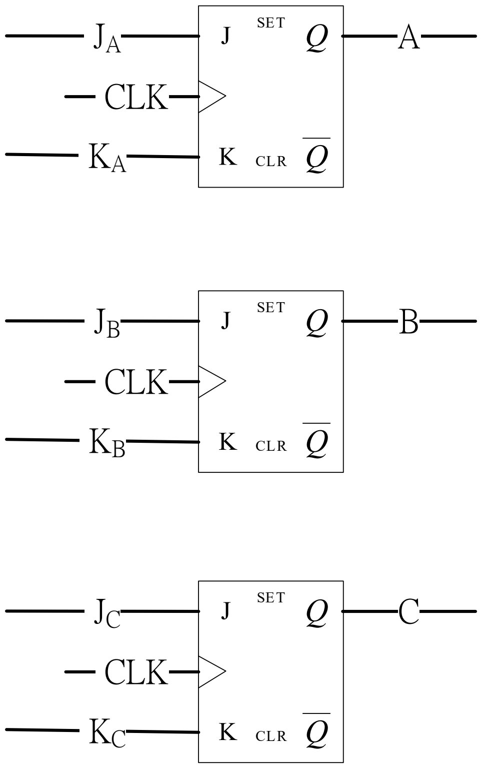
XA \ BC	00	01	11	10
00	X	1	1	X
01	X	1	1	X
11	X	1	1	X
10	X	1	1	X

$K_C = 1$

$J_C = K_C$

5.8 Design Procedure (22/29)
Analysis with *JK* Flip-Flops: Design an FSM Circuit (3)

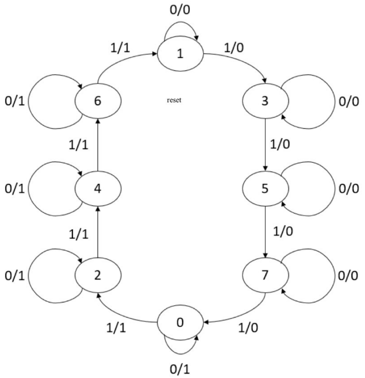
Problem => State diagram => State table => Equation => Circuit



5.8 Design Procedure (22/28)
Analysis with *JK* Flip-Flops: Example 4

Problem → State diagram → State table → Equation → Circuit

- 設計具有3個JK型正反器A, B, C、1個輸入X與1個輸出Y之3-bit counter。
  - $X = 0$ 時，狀態值保持不變
  - $X = 1$ 時，
    - 目前狀態值為奇數時，
      - 若目前狀態值為7，次一狀態值為0。
      - 否則，狀態值加2。
    - 目前狀態值為偶數時，
      - 若目前狀態值為6，次一狀態值為1。
      - 否則，狀態值加2。
  - 目前狀態值為奇數時，$Y = 0$；為偶數時，$Y = 1$



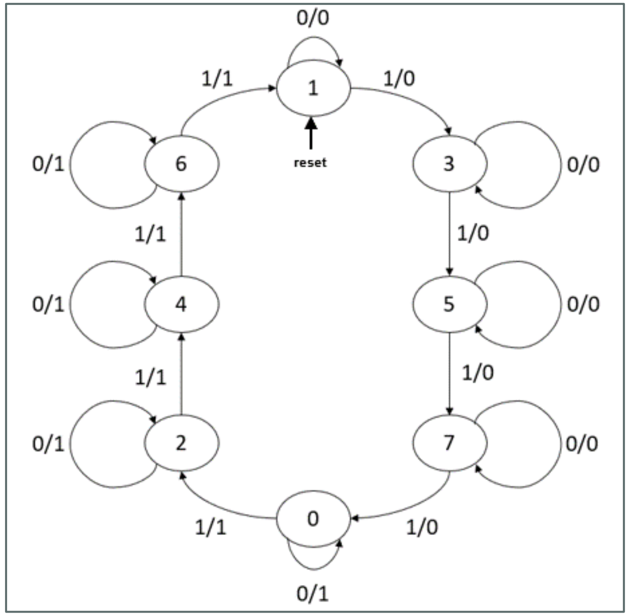
State diagram

5.8 Design Procedure (24/29)
Analysis with *JK* Flip-Flops: Design an FSM Circuit (4)

JK正反器特徵表

Q(t)	Q(t+1)	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Problem => State diagram => State table => Equation => Circuit



State diagram

Present State				Next State				Next State JK					
X	A	B	C	A	B	C	Y	JA	KA	JB	KB	JC	KC
0	0	0	0	0	0	0	1	0	X	0	X	0	X
0	0	0	1	0	0	1	0	0	X	0	X	X	0
0	0	1	0	0	1	0	1	0	X	X	0	0	X
0	0	1	1	0	1	1	0	0	X	X	0	X	0
0	1	0	0	1	0	0	1	X	0	0	X	0	X
0	1	0	1	1	0	1	0	X	0	0	X	X	0
0	1	1	0	1	1	0	1	X	0	X	0	0	X
0	1	1	1	1	1	1	0	X	0	X	0	X	0
1	0	0	0	0	1	0	1	0	X	1	X	0	X
1	0	0	1	0	1	1	0	0	X	1	X	X	0
1	0	1	0	1	0	0	1	1	X	X	0	0	X
1	0	1	1	1	0	1	0	1	X	X	0	X	0
1	1	0	0	1	1	0	1	X	0	1	X	0	X
1	1	0	1	1	1	1	0	X	0	1	X	X	0
1	1	1	0	0	0	1	1	X	1	X	0	1	X
1	1	1	1	0	0	0	0	X	1	X	0	X	1

State table

5.8 Design Procedure (24/28)
Analysis with *JK* Flip-Flops: Example 4

Problem → State diagram → State table → Equation → Circuit

Present State				Next State				Flip-Flop Inputs					
X	A	B	C	A	B	C	Y	JA	KA	JB	KB	JC	KC
0	0	0	0	0	0	0	1	0	X	0	X	0	X
0	0	0	1	0	0	1	0	0	X	0	X	X	0
0	0	1	0	0	1	0	1	0	X	X	0	0	X
0	0	1	1	0	1	1	0	0	X	X	0	X	0
0	1	0	0	1	0	0	1	X	0	0	X	0	X
0	1	0	1	1	0	1	0	X	0	0	X	X	0
0	1	1	0	1	1	0	1	X	0	X	0	0	X
0	1	1	1	1	1	1	0	X	0	X	0	X	0
1	0	0	0	0	1	0	1	0	X	1	X	0	X
1	0	0	1	0	1	1	0	0	X	1	X	X	0
1	0	1	0	1	0	0	1	1	X	X	0	0	X
1	0	1	1	1	0	1	0	1	X	X	0	X	0
1	1	0	0	1	1	0	1	X	0	1	X	0	X
1	1	0	1	1	1	1	0	X	0	1	X	X	0
1	1	1	0	0	0	1	1	X	1	X	0	1	X
1	1	1	1	0	0	0	0	X	1	X	0	X	1

State table

XA \ BC	00	01	11	10
00				
01	X	X	X	X
11	X	X	X	X
10			1	1

$J_A = XB$

XA \ BC	00	01	11	10
00	X	X	X	X
01				
11			1	1
10	X	X	X	X

$K_A = XB$

5.8 Design Procedure (25/28)
Analysis with *JK* Flip-Flops: Example 4

Problem → State diagram → State table → Equation → Circuit

Present State				Next State				Flip-Flop Inputs					
X	A	B	C	A	B	C	Y	JA	KA	JB	KB	JC	KC
0	0	0	0	0	0	0	1	0	X	0	X	0	X
0	0	0	1	0	0	1	0	0	X	0	X	X	0
0	0	1	0	0	1	0	1	0	X	X	0	0	X
0	0	1	1	0	1	1	0	0	X	X	0	X	0
0	1	0	0	1	0	0	1	X	0	0	X	0	X
0	1	0	1	1	0	1	0	X	0	0	X	X	0
0	1	1	0	1	1	0	1	X	0	X	0	0	X
0	1	1	1	1	1	1	0	X	0	X	0	X	0
1	0	0	0	0	1	0	1	0	X	1	X	0	X
1	0	0	1	0	1	1	0	0	X	1	X	X	0
1	0	1	0	1	0	0	1	1	X	X	0	0	X
1	0	1	1	1	0	1	0	1	X	X	0	X	0
1	1	0	0	1	1	0	1	X	0	1	X	0	X
1	1	0	1	1	1	1	0	X	0	1	X	X	0
1	1	1	0	0	0	1	1	X	1	X	0	1	X
1	1	1	1	0	0	0	0	X	1	X	0	X	1

State table

XA \ BC	00	01	11	10
00			X	X
01			X	X
11	1	1	X	X
10	1	1	X	X

$J_B = X$

XA \ BC	00	01	11	10
00	X	X	0	0
01	X	X	0	0
11	X	X	0	0
10	X	X	0	0

$K_B = 0$

5.8 Design Procedure (26/28)
Analysis with *JK* Flip-Flops: Example 4

Problem → State diagram → State table → Equation → Circuit

Present State				Next State				Flip-Flop Inputs					
X	A	B	C	A	B	C	Y	JA	KA	JB	KB	JC	KC
0	0	0	0	0	0	0	1	0	X	0	X	0	X
0	0	0	1	0	0	1	0	0	X	0	X	X	0
0	0	1	0	0	1	0	1	0	X	X	0	0	X
0	0	1	1	0	1	1	0	0	X	X	0	X	0
0	1	0	0	1	0	0	1	X	0	0	X	0	X
0	1	0	1	1	0	1	0	X	0	0	X	X	0
0	1	1	0	1	1	0	1	X	0	X	0	0	X
0	1	1	1	1	1	1	0	X	0	X	0	X	0
1	0	0	0	0	1	0	1	0	X	1	X	0	X
1	0	0	1	0	1	1	0	0	X	1	X	X	0
1	0	1	0	1	0	0	1	1	X	X	0	0	X
1	0	1	1	1	0	1	0	1	X	X	0	X	0
1	1	0	0	1	1	0	1	X	0	1	X	0	X
1	1	0	1	1	1	1	0	X	0	1	X	X	0
1	1	1	0	0	0	1	1	X	1	X	0	1	X
1	1	1	1	0	0	0	0	X	1	X	0	X	1

State table

XA \ BC	00	01	11	10
00		X	X	
01		X	X	
11		X	X	1
10		X	X	

$J_C = XAB$

XA \ BC	00	01	11	10
00	X			X
01	X			X
11	X		1	X
10	X			X

$K_C = XAB$

5.8 Design Procedure (27/28)
Analysis with *JK* Flip-Flops: Example 4

Problem → State diagram → State table → Equation → Circuit

Present State				Next State				Flip-Flop Inputs					
X	A	B	C	A	B	C	Y	JA	KA	JB	KB	JC	KC
0	0	0	0	0	0	0	1	0	X	0	X	0	X
0	0	0	1	0	0	1	0	0	X	0	X	X	0
0	0	1	0	0	1	0	1	0	X	X	0	0	X
0	0	1	1	0	1	1	0	0	X	X	0	X	0
0	1	0	0	1	0	0	1	X	0	0	X	0	X
0	1	0	1	1	0	1	0	X	0	0	X	X	0
0	1	1	0	1	1	0	1	X	0	X	0	0	X
0	1	1	1	1	1	1	0	X	0	X	0	X	0
1	0	0	0	0	1	0	1	0	X	1	X	0	X
1	0	0	1	0	1	1	0	0	X	1	X	X	0
1	0	1	0	1	0	0	1	1	X	X	0	0	X
1	0	1	1	1	0	1	0	1	X	X	0	X	0
1	1	0	0	1	1	0	1	X	0	1	X	0	X
1	1	0	1	1	1	1	0	X	0	1	X	X	0
1	1	1	0	0	0	1	1	X	1	X	0	1	X
1	1	1	1	0	0	0	0	X	1	X	0	X	1

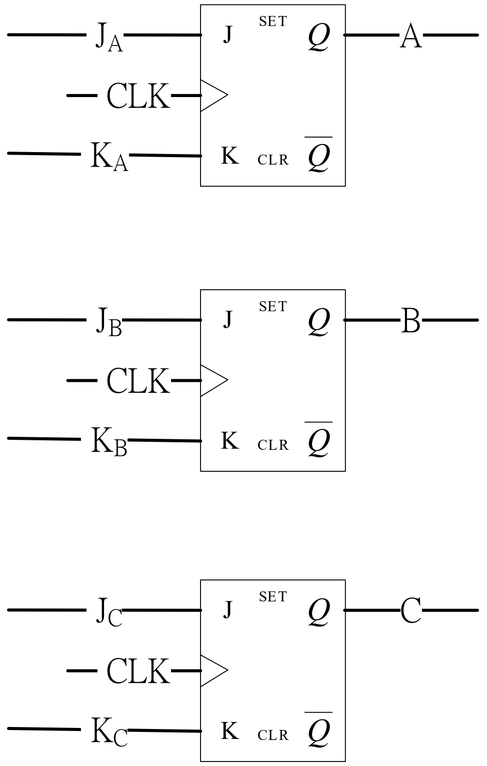
State table

XA \ BC	00	01	11	10
00	1			1
01	1			1
11	1			1
10	1			1

$Y = C'$

5.8 Design Procedure (29/29)
Analysis with *JK* Flip-Flops: Design an FSM Circuit (4)

Problem => State diagram => State table => Equation => Circuit



$Y =$